

《数字信号处理》课程讨论课

适用于时变信号的时频分析方法

提 纲

1. 课堂讨论课的目标及意义

2. 需要理解的内容

- 具有时间局部化性质的加窗（短时）**Fourier**分析
- 具有变时频分辨能力的小波分析、**S**变换方法

3. 要求

1. 目标及意义

在深入理解经典**Fourier**变换的基础上，通过较系统介绍加窗**Fourier**变换（ **Windowed Fourier Transform, WFT** ）、小波变换（**Wavelet transform, WT**）等定义，理解时间分辨率、频率分辨率及时、频局部化性质等基本概念；通过编程实现**WFT**等时频分析算法，理解时变信号中时频结构等信息。

时频分析理论简介

傅里叶变换(FT)

- (1) 对信号做全局变换,不能对信号进行局部刻画;
- (2) 适合于分析平稳信号;

短时傅里叶变换(WFT)

- (1) 有一定的局部刻画能力,但是分辨率单一,不能兼顾时频分辨率的需求;
- (2) 可用于非平稳信号分析;

小波变换(WT)

- (1) 同时兼顾信号对时间和频率分辨率的需求;
- (2) 适合于分析非平稳信号.

提 纲

1. 课堂讨论课的目标及意义

2. 需要理解的内容

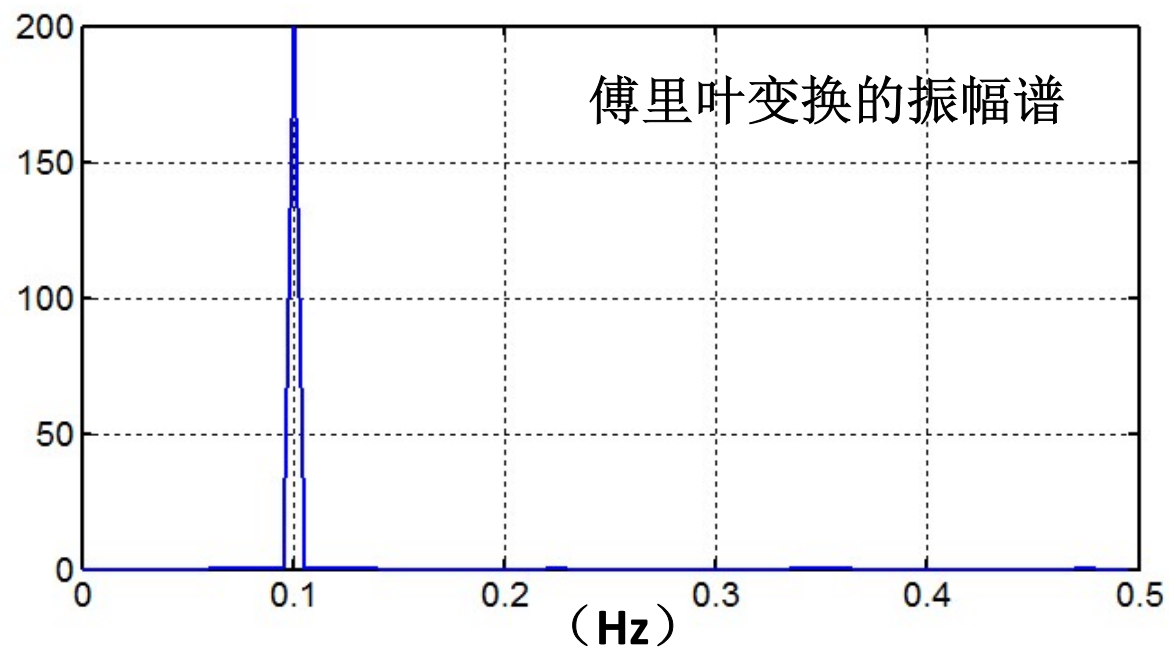
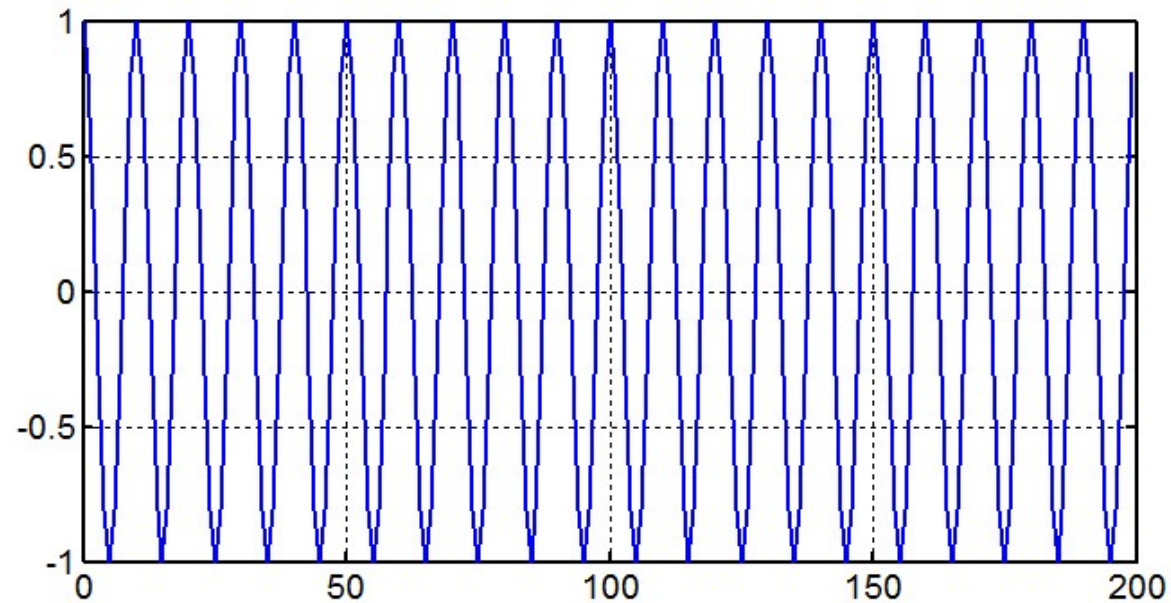
- 具有时间局部化性质的加窗（短时）**Fourier**分析
- 具有变时频分辨能力的小波分析、 **S**变换方法

3. 要求

傅里叶变换

Fourier Transform

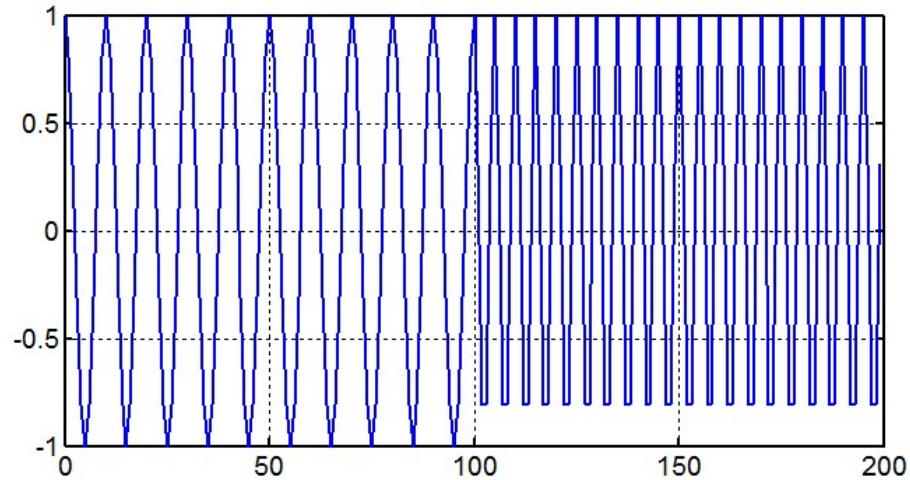
0.1Hz合成信号



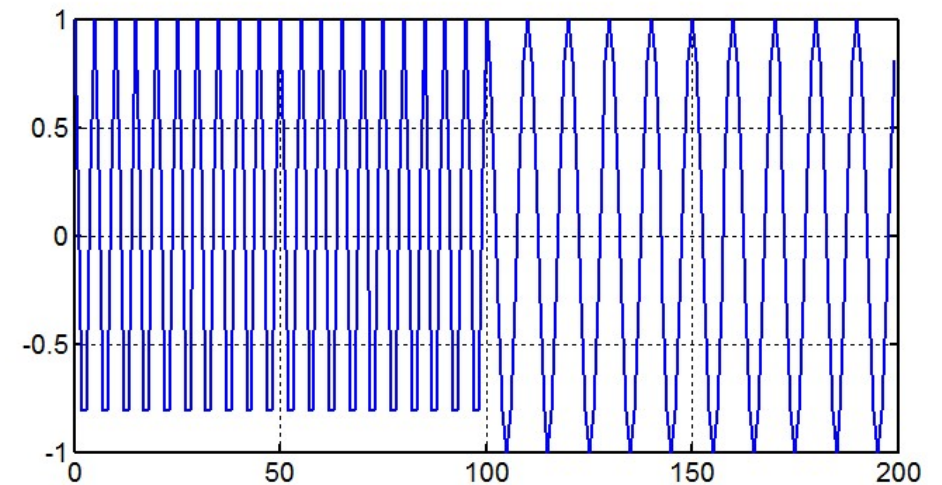
傅里叶变换

Fourier Transform

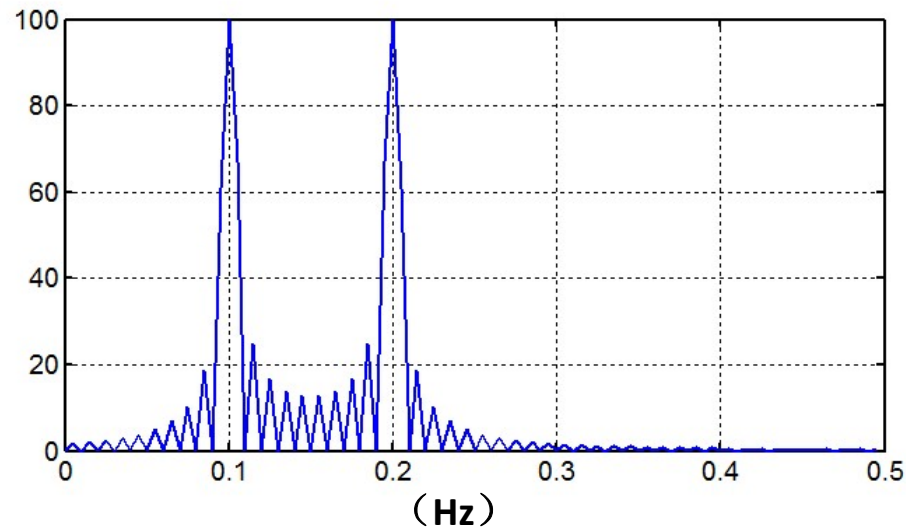
0.1Hz级联0.2Hz合成信号



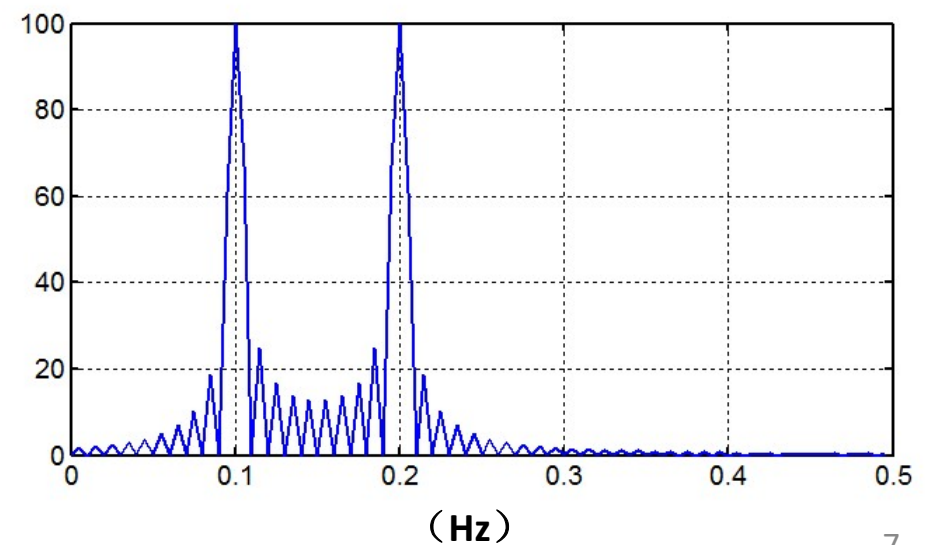
0.2Hz级联0.1Hz合成信号



0.1Hz级联0.2Hz合成信号傅里叶振幅谱



0.2Hz级联0.1Hz合成信号傅里叶振幅谱



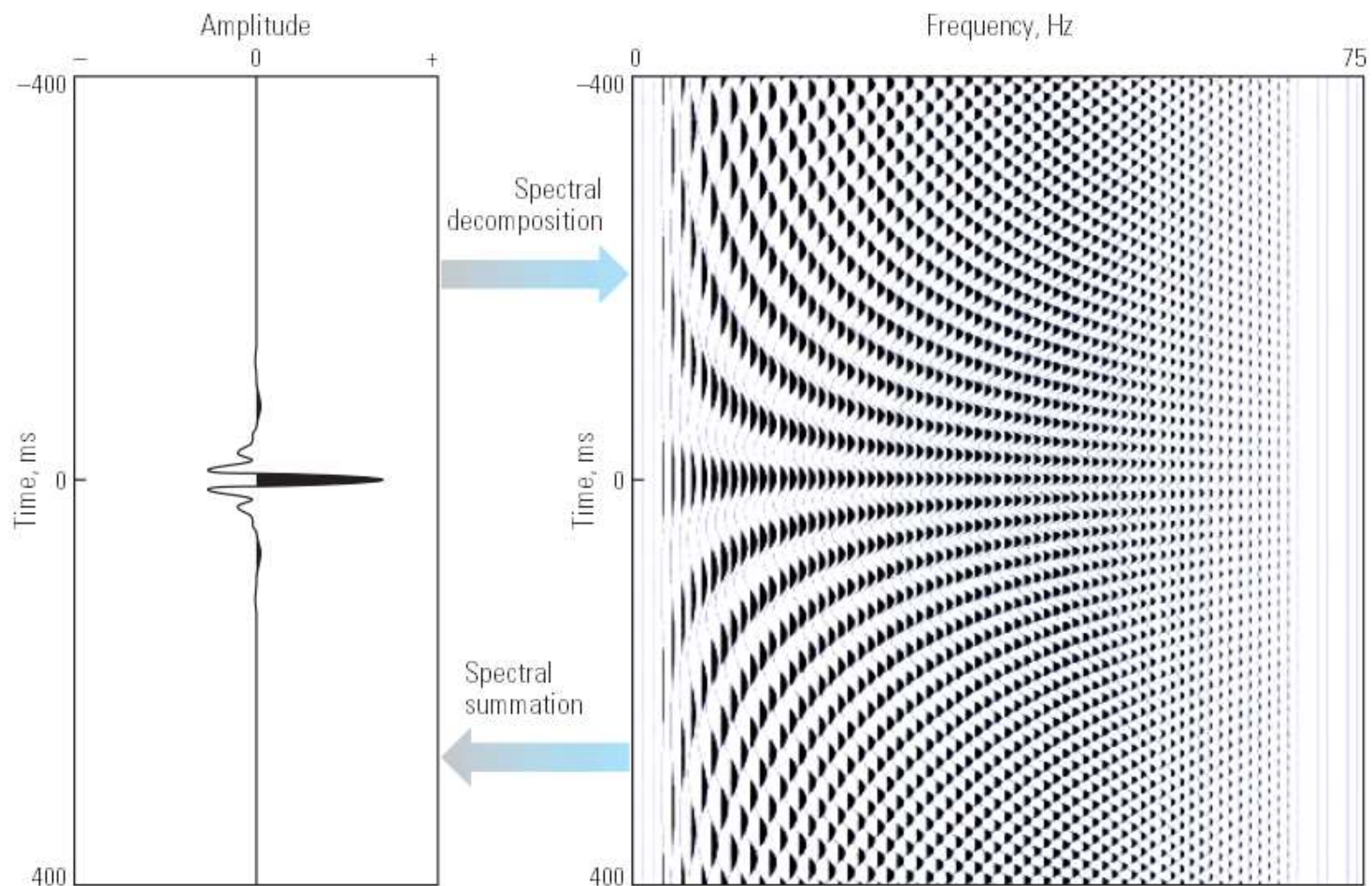
具有时间局部化性质的加窗Fourier分析

Fourier正反变换公式

$$S(\xi) = \int_{-\infty}^{+\infty} s(t) e^{-j2\pi\xi t} dt$$

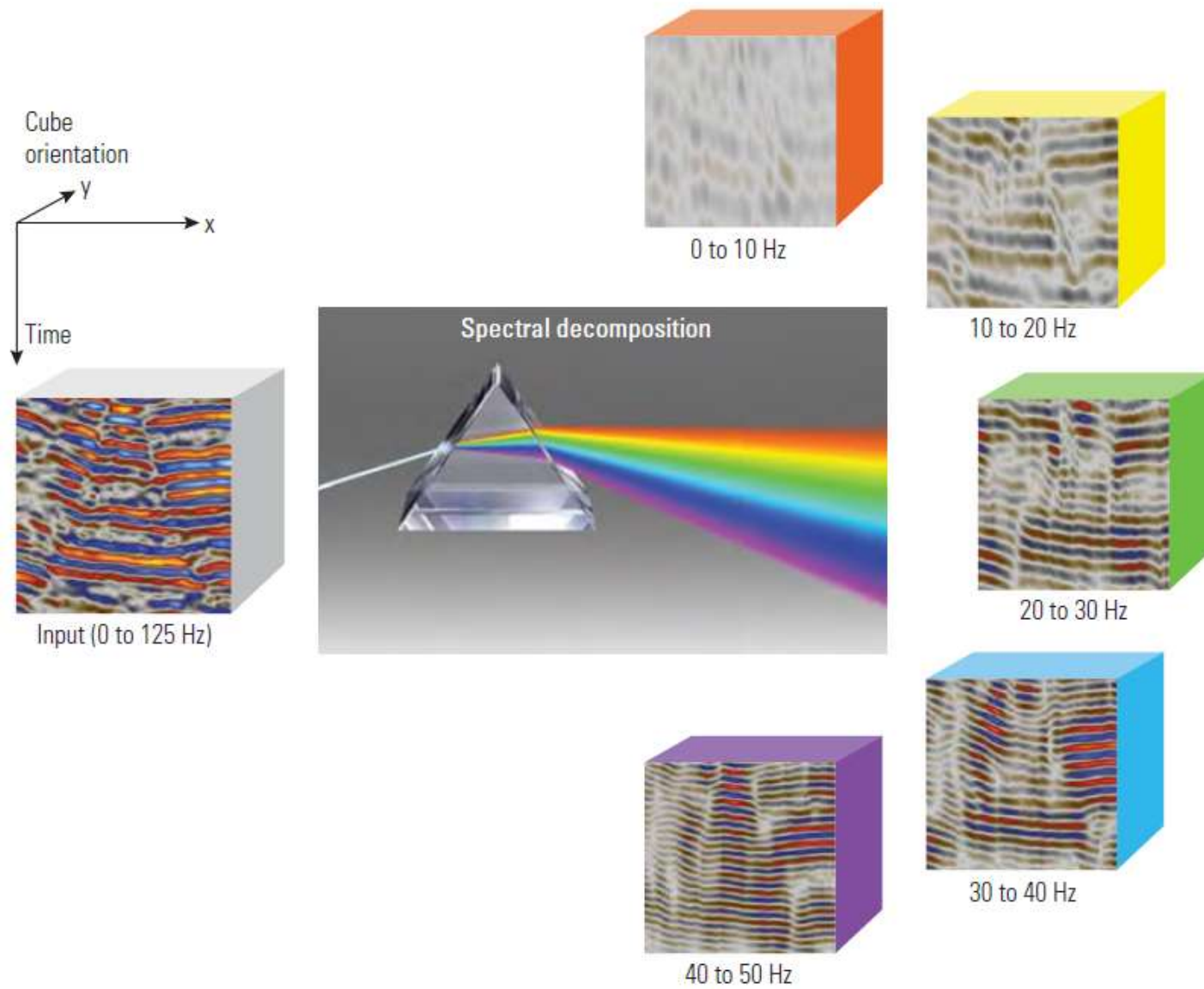
$$s(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} S(\xi) e^{j2\pi\xi t} d\xi$$

宽带信号的Fourier成分



(Aarre et al., 2012)

宽带信号的Fourier成分

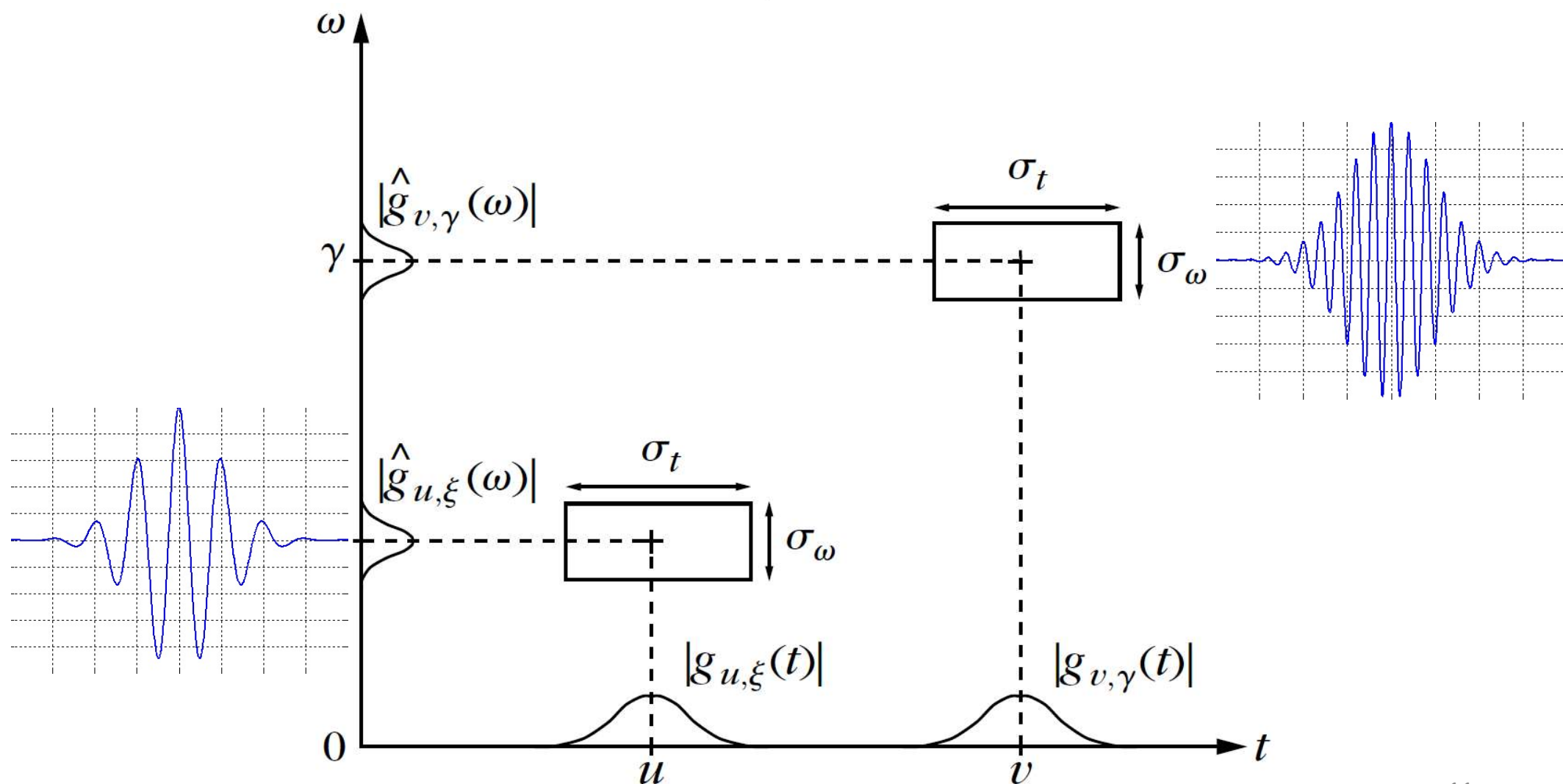


(Aarre et al., 2012)

加窗傅里叶变换

Windowed Fourier Transform

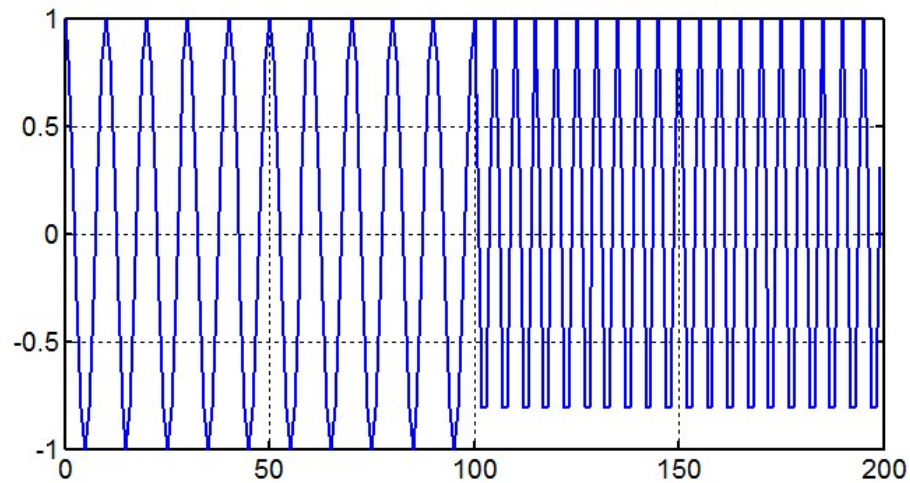
$$Sf(u, \xi) = \langle f, g_{u, \xi} \rangle = \int_{-\infty}^{+\infty} f(t) g(t - u) e^{-i\xi t} dt$$



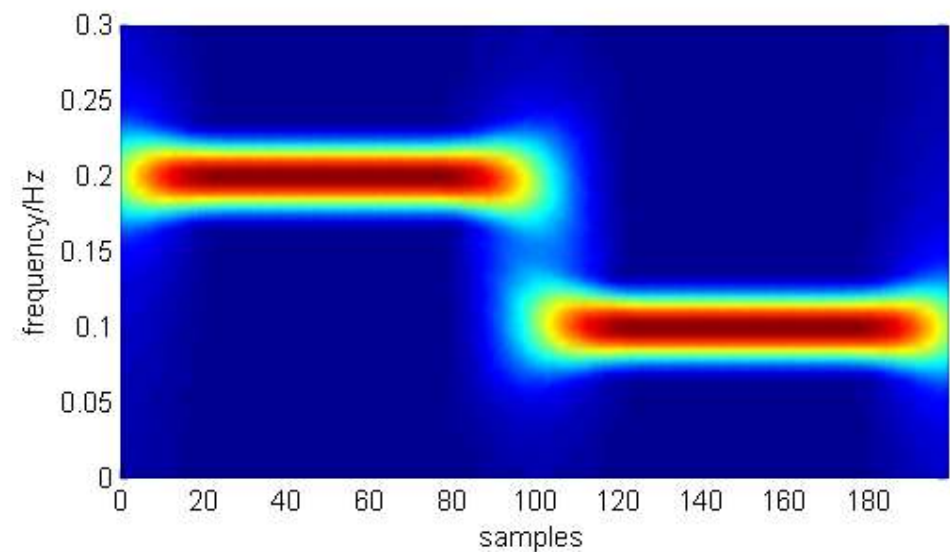
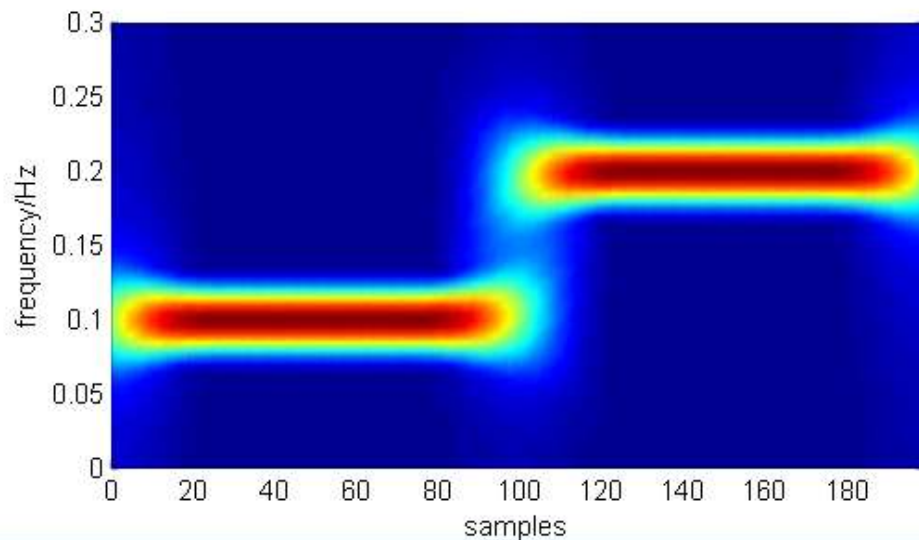
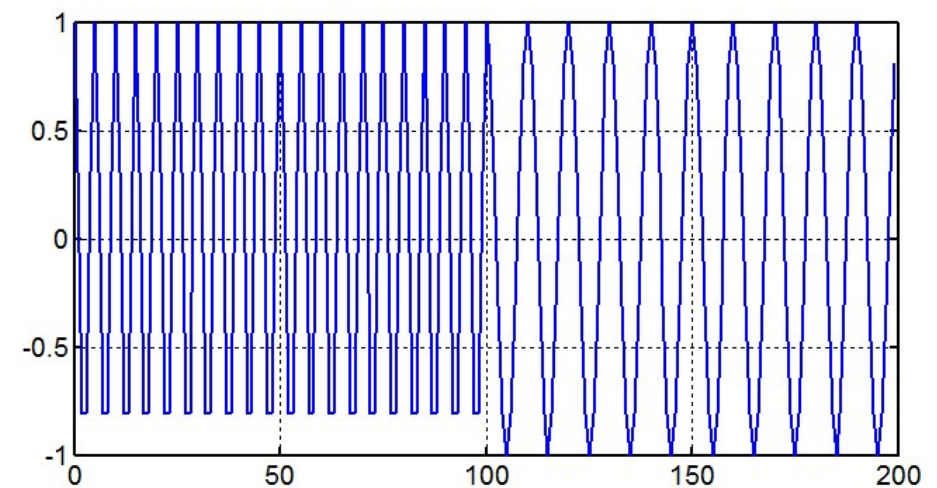
加窗傅里叶变换

Windowed Fourier Transform

0.1Hz级联0.2Hz合成信号



0.2Hz级联0.1Hz合成信号



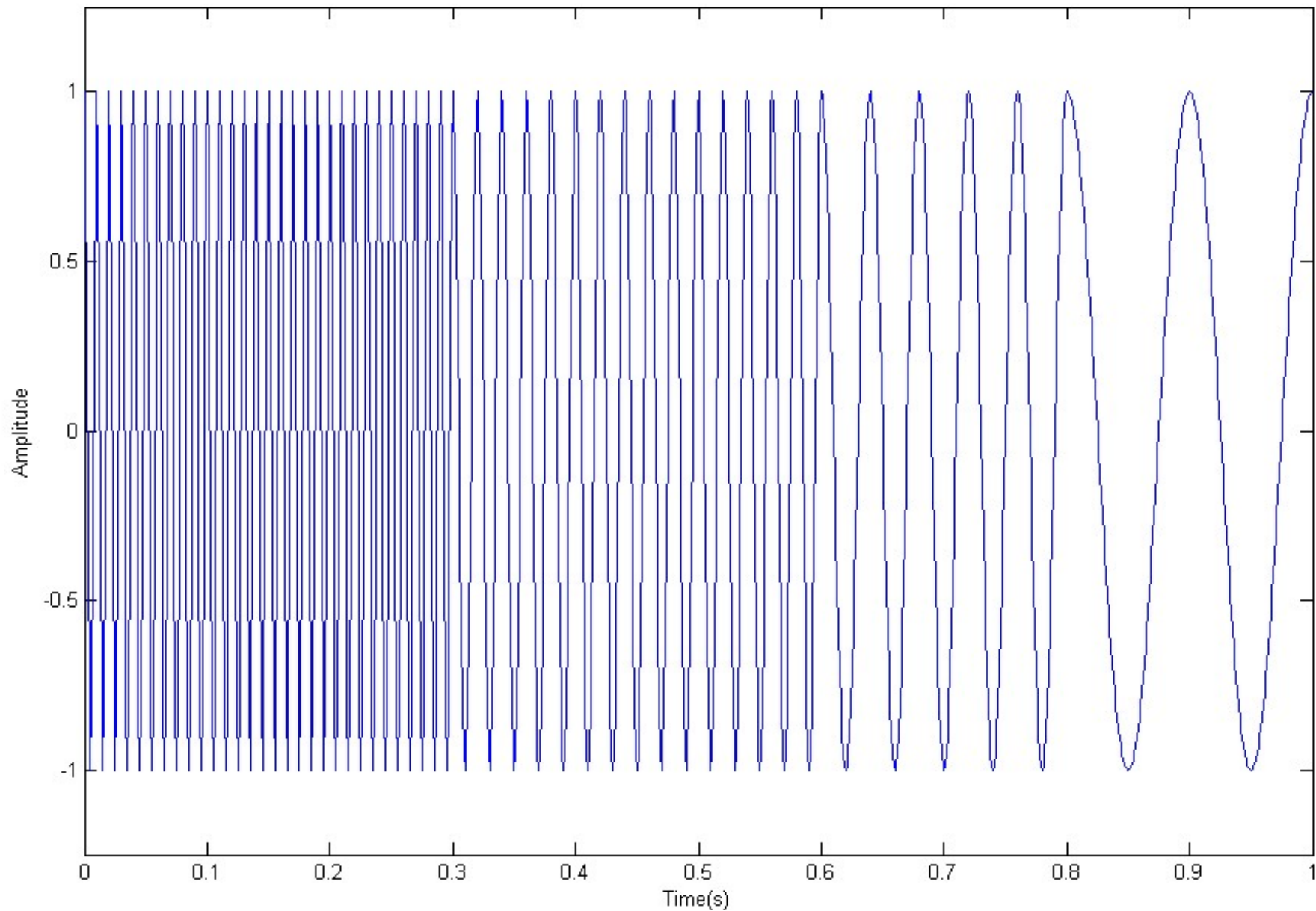


图 a, 其中0-0.3s为100Hz正弦信号, 0.3-0.6s为50Hz正弦信号,0.6-0.8s为25Hz正弦信号,0.8-1s为10Hz正弦信号.

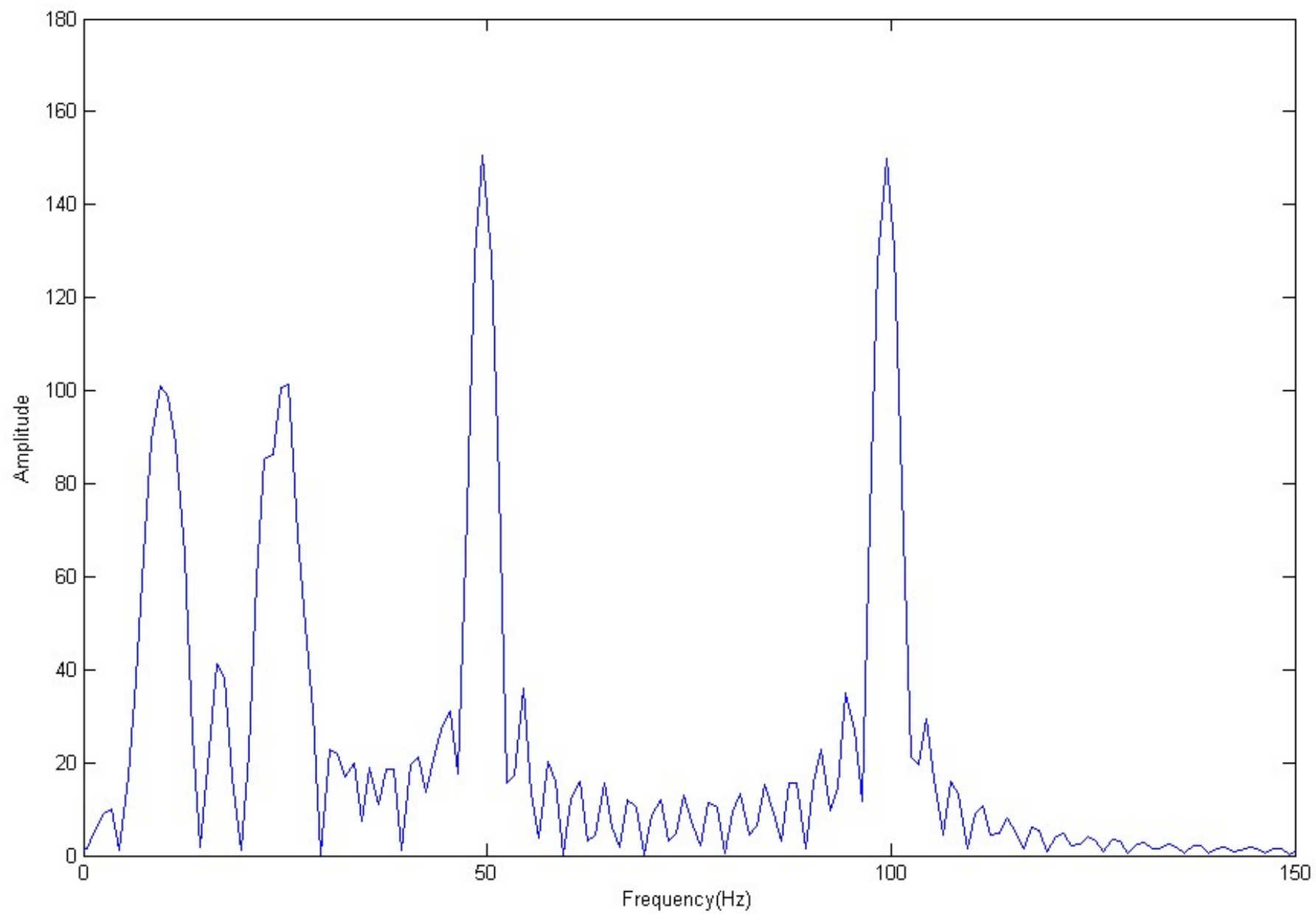


图 b, Fourier振幅谱

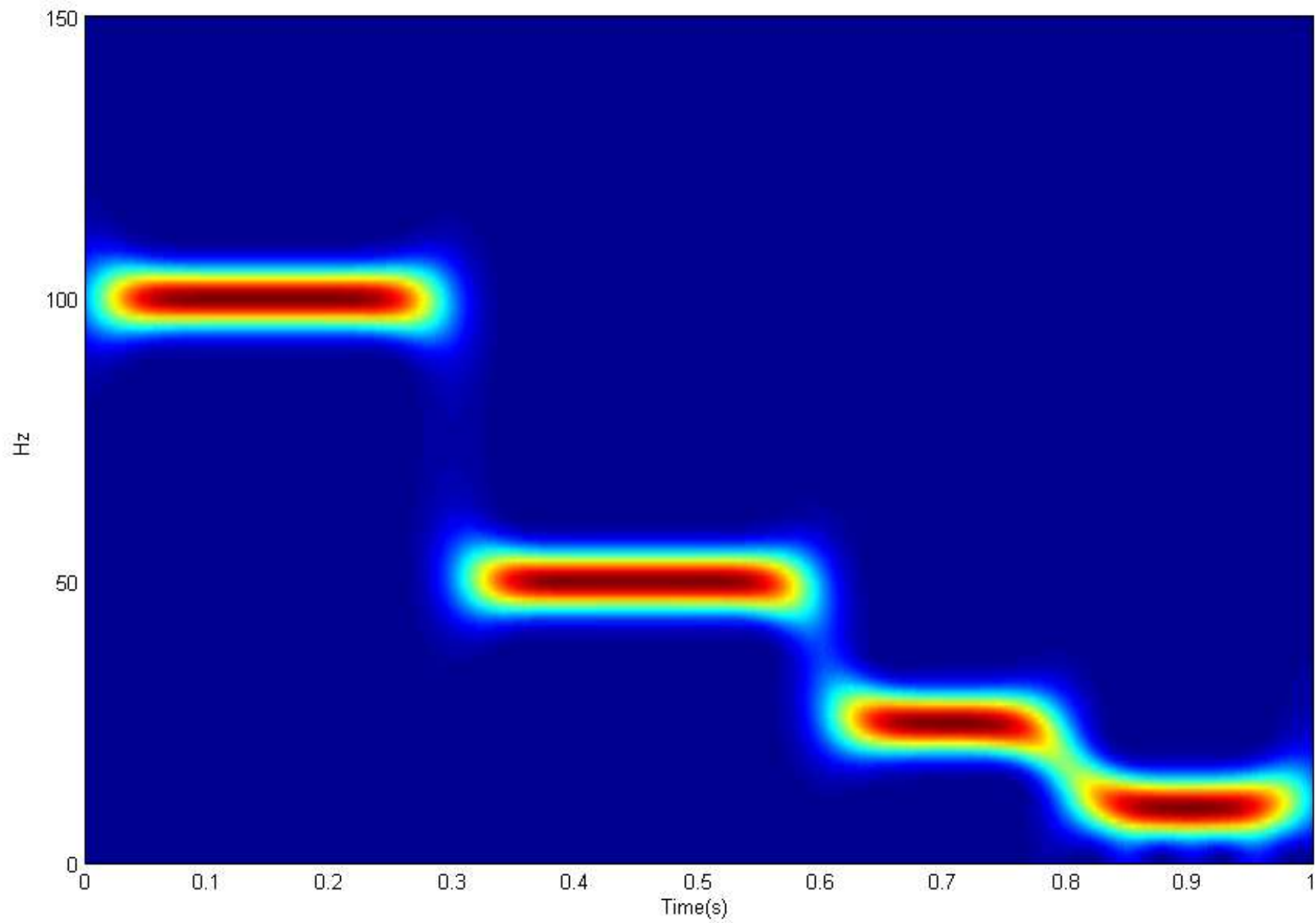
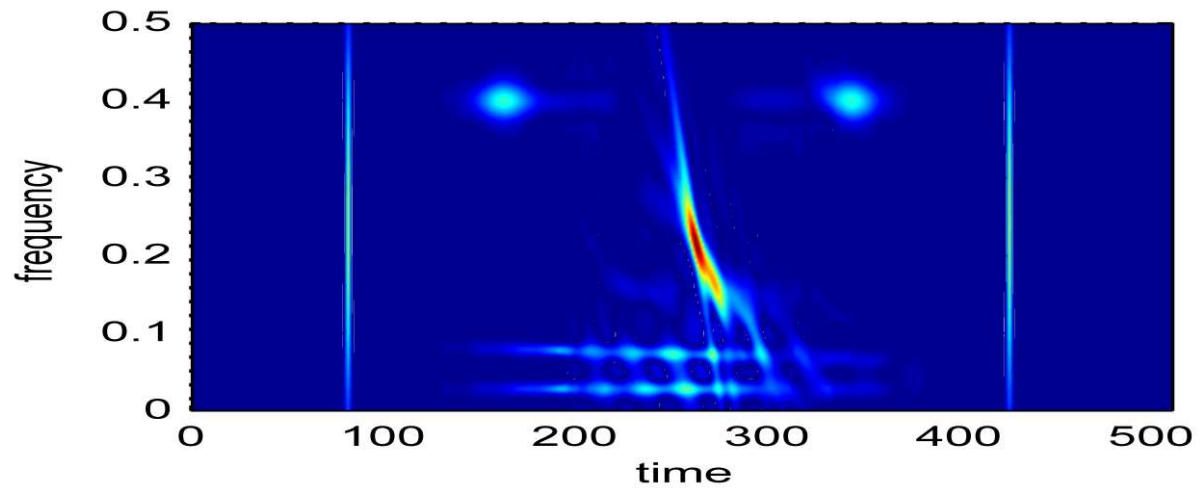
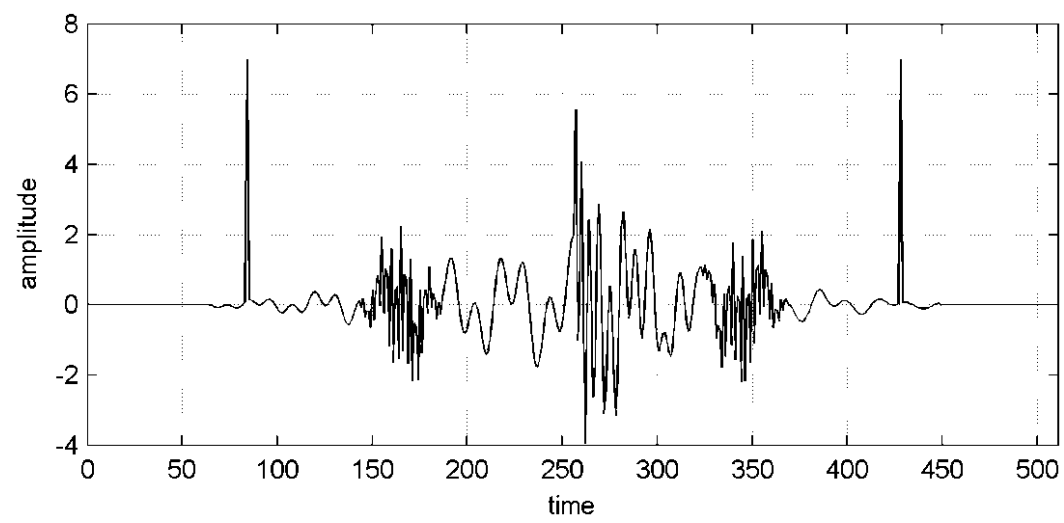
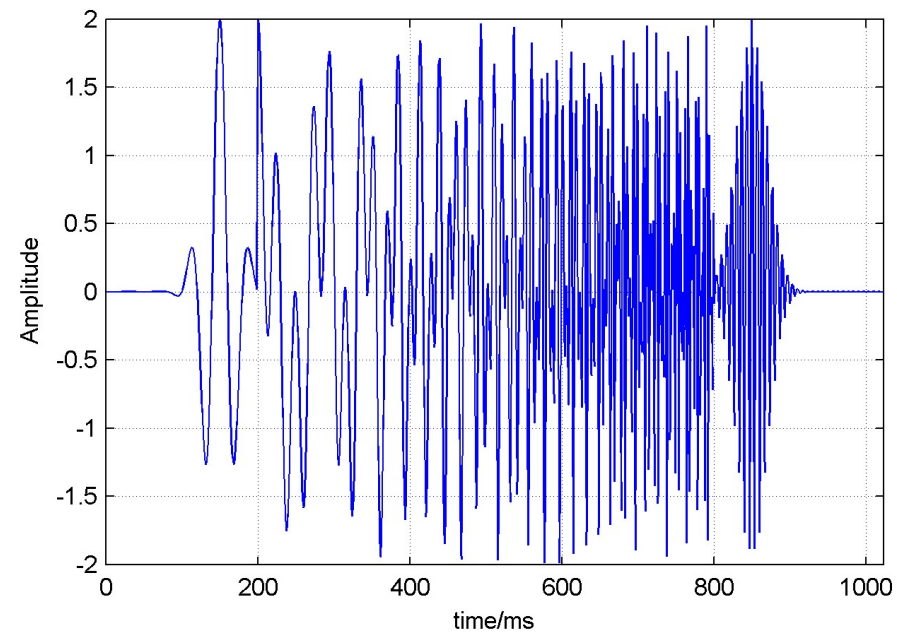


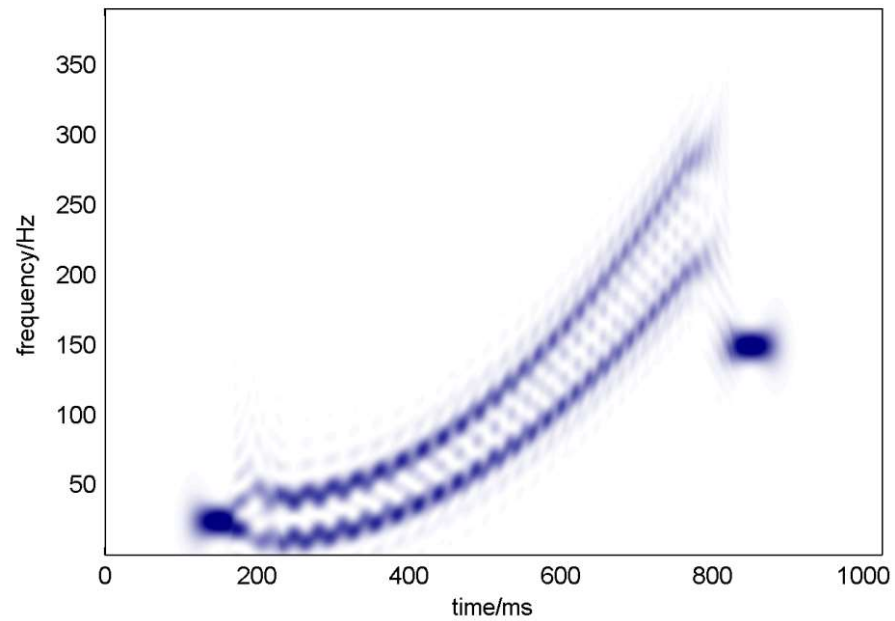
图 d, 较窄的Hanning窗对图a的短时Fourier变换



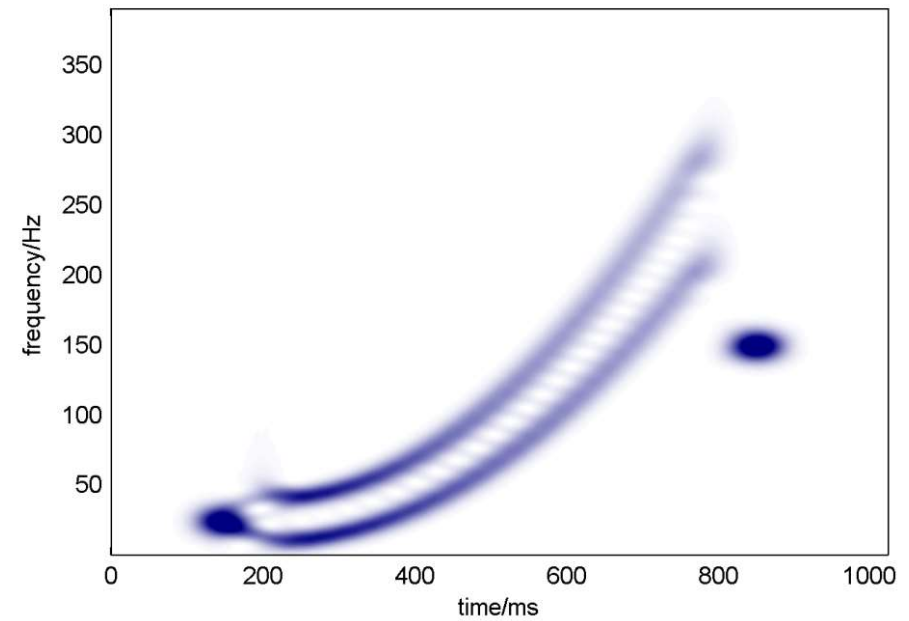
上图为 测试信号，下图为测试信号的时频图.



测试信号



64点高斯窗短时Fourier变换结果.



128点高斯窗短时Fourier变换结果. 17

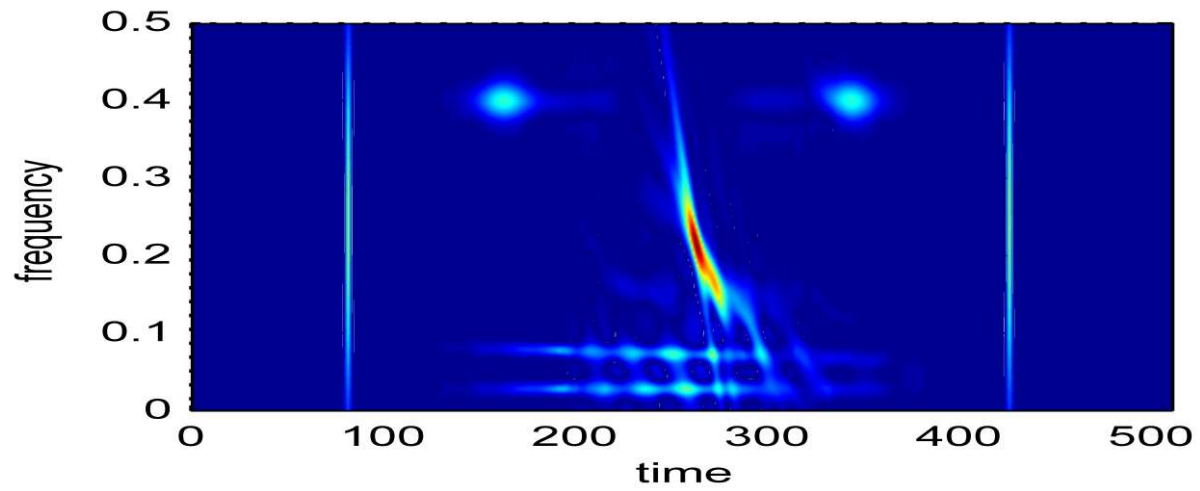
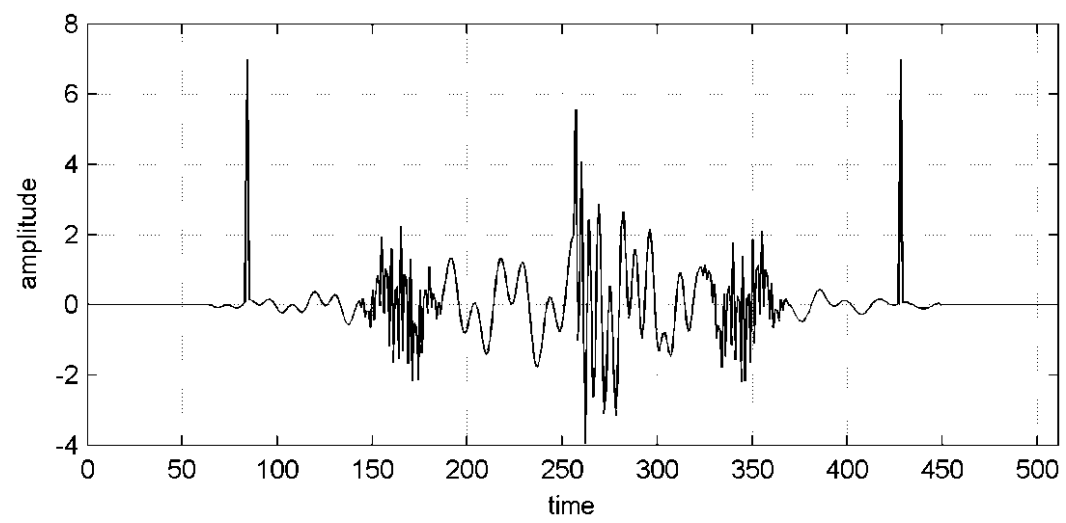
加窗Fourier变换的定义

$$Sf(u, \xi) = \int_{-\infty}^{+\infty} f(t) g^*(u-t) e^{-j2\pi\xi t} dt$$

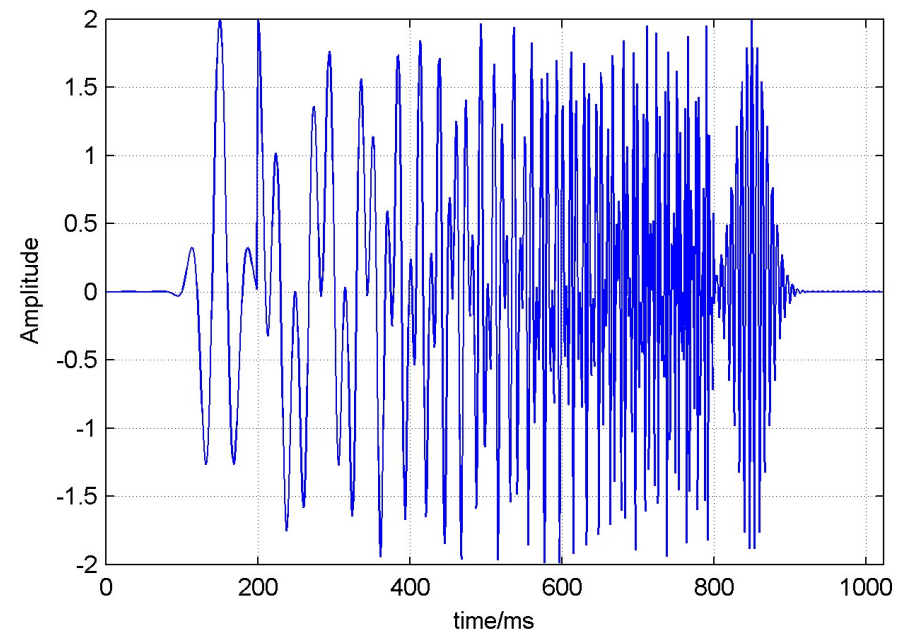
$$f'(t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} Sf(u, \xi) \gamma(u-t) e^{j2\pi\xi t} du d\xi$$

→ $f'(t) = f(t), \quad \forall t$ 完全信号重构（恒等分辨）
Perfect Signal Reconstruction

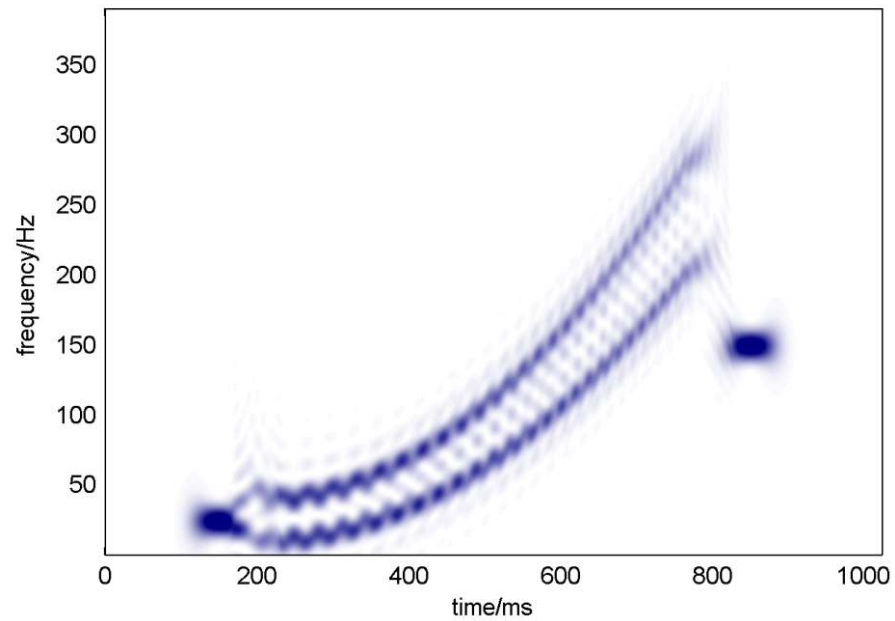
完全信号重构条件： $\int_{-\infty}^{\infty} g(t) \gamma^*(t) dt = 1$



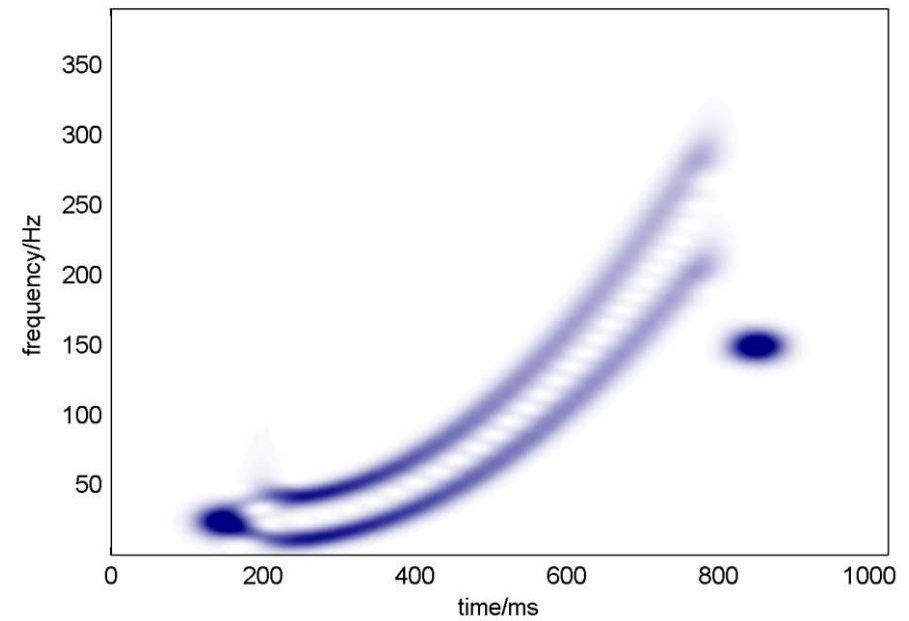
上图为 测试信号，下图为测试信号的时频图.



测试信号



64点高斯窗短时Fourier变换结果.



128点高斯窗短时Fourier变换结果. 20

小波变换的发展历史简介

1909: Alfred Haar——发现了Haar小波

1980年 Morlet——Morlet小波

1986年 Y.Meyer——提出了第一个正交小波Meyer小波

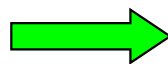
1988年 S. Mallat——Mallat快速算法塔式分解和重构算法

连续小波变换定义的定义

$$W(u, s) = \int_{-\infty}^{+\infty} f(t) \frac{1}{\sqrt{|s|}} \psi^* \left(\frac{t-u}{s} \right) dt$$

小波函数需满足可容许性条件:

$$C_{\psi} = \int_{-\infty}^{+\infty} \frac{|\psi(\omega)|^2}{|\omega|} d\omega < \infty$$

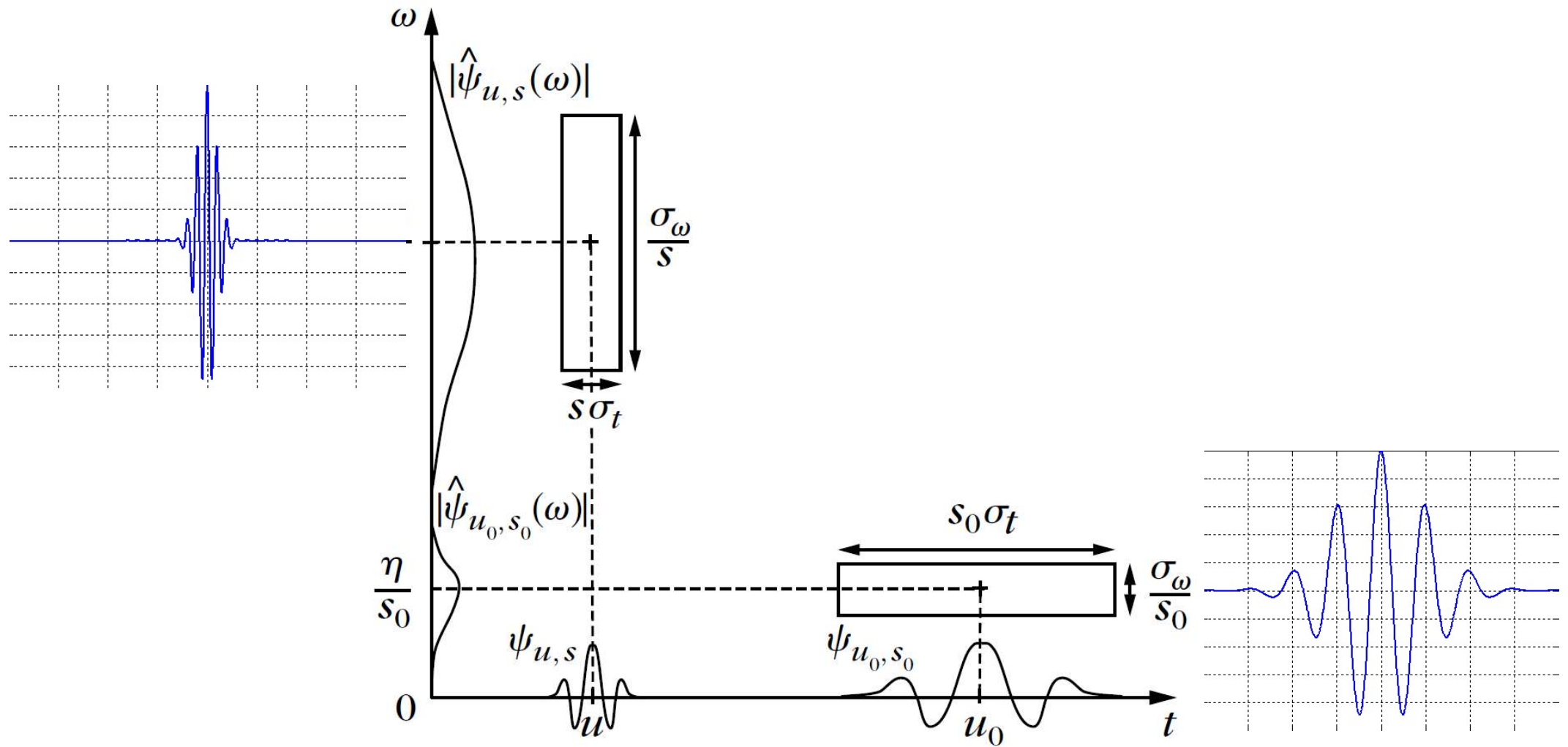


当 $\omega = 0$ 时, $\psi(\omega) = 0$;

$$\int_{-\infty}^{+\infty} \psi(t) dt = 0;$$

一维连续小波变换

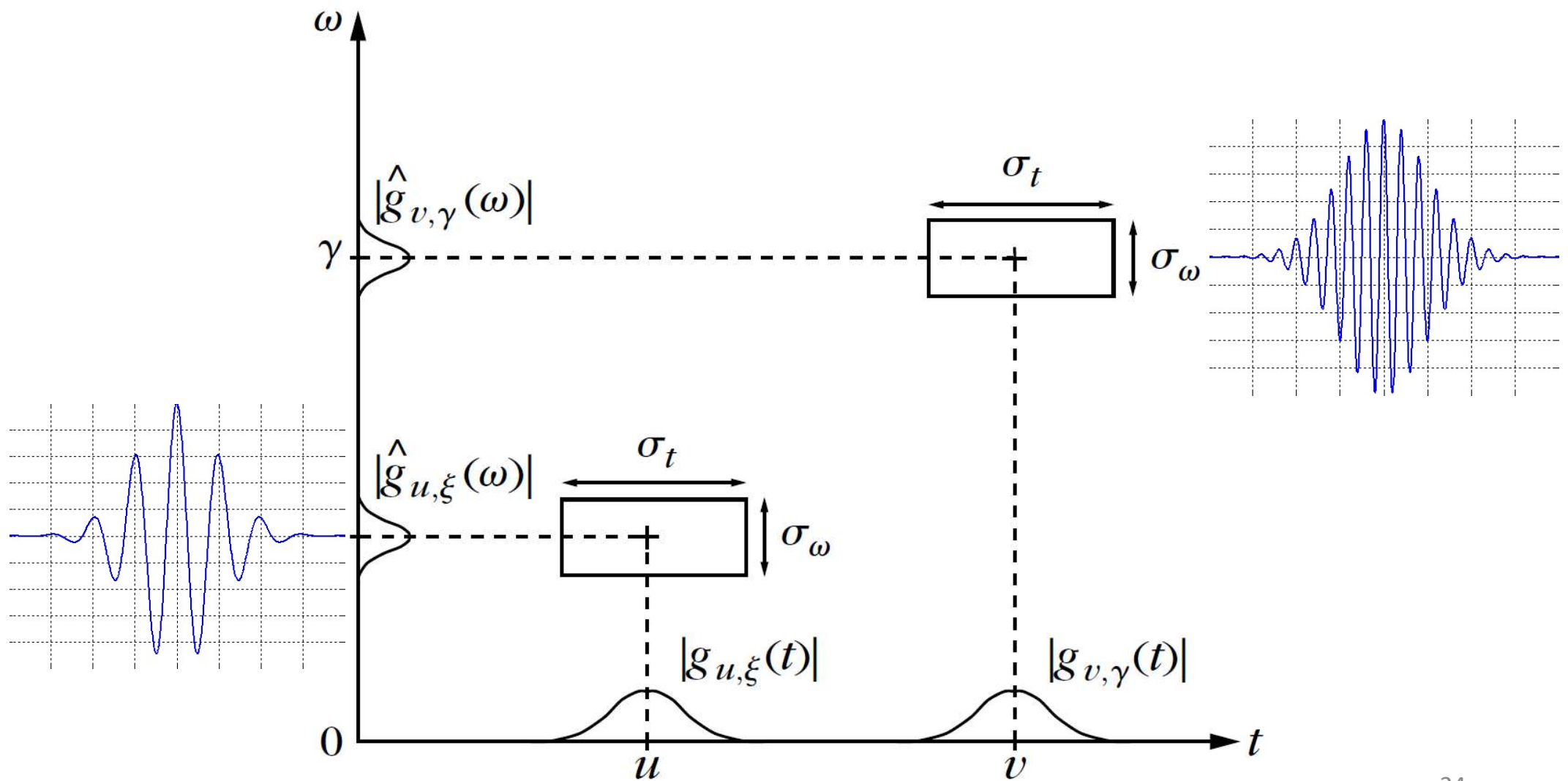
1D Continuous Wavelet Transform



加窗傅里叶变换

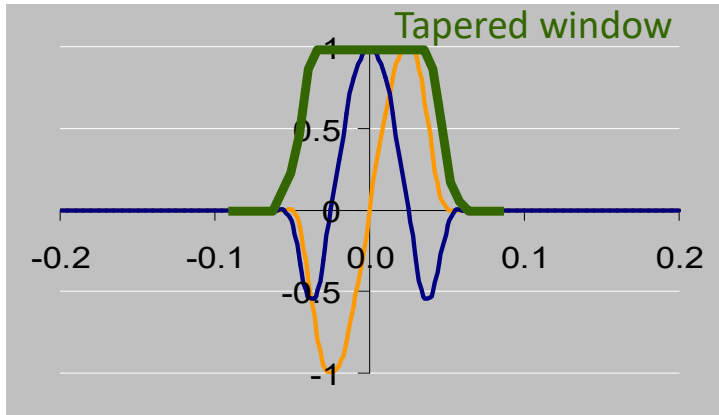
Windowed Fourier Transform

$$Sf(u, \xi) = \langle f, g_{u, \xi} \rangle = \int_{-\infty}^{+\infty} f(t) g(t - u) e^{-i\xi t} dt$$

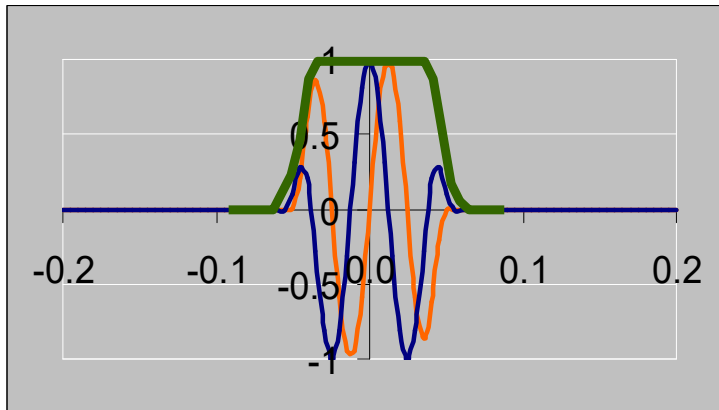


WFT wavelets

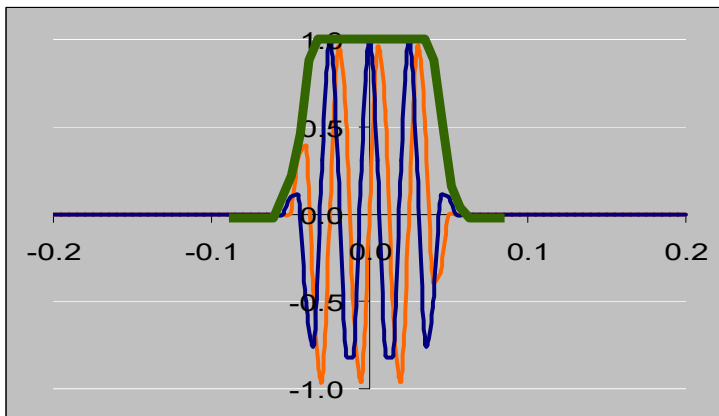
$f_c = 10$ Hz



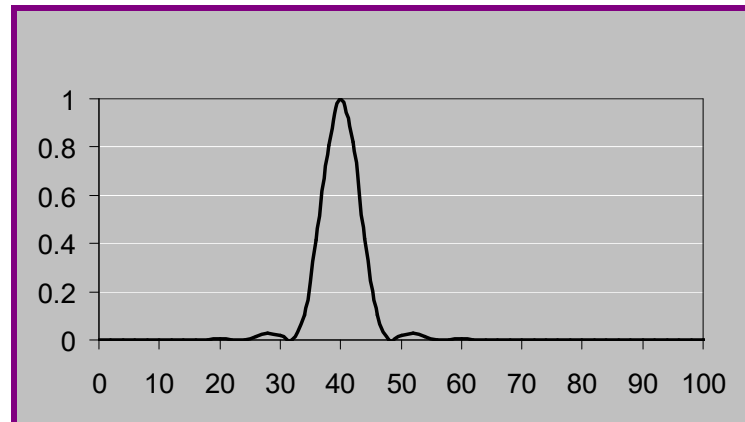
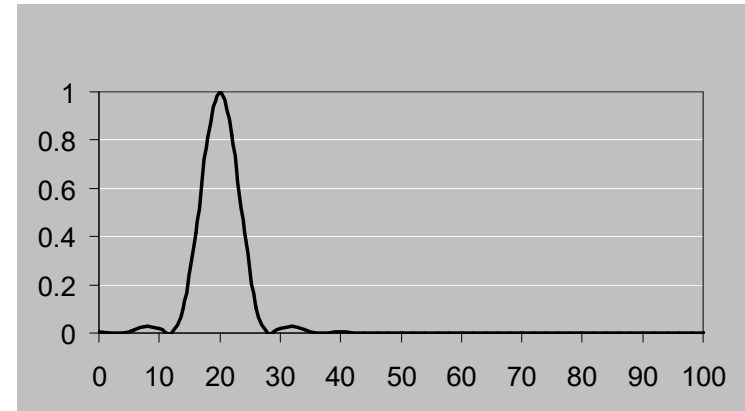
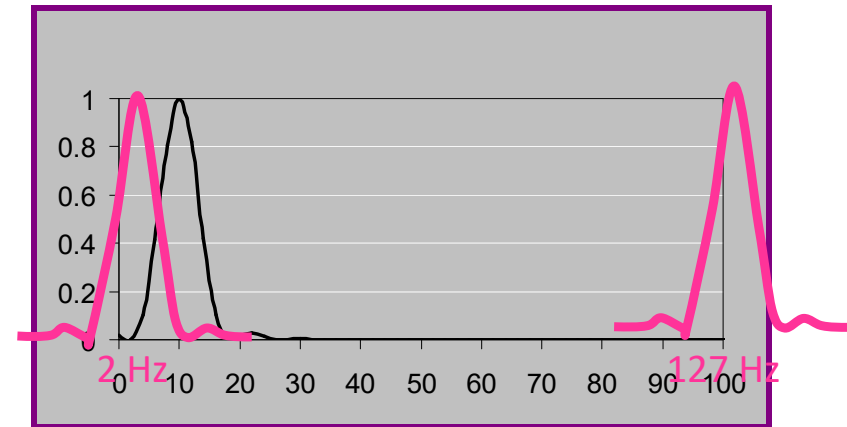
$f_c = 20$ Hz



$f_c = 40$ Hz



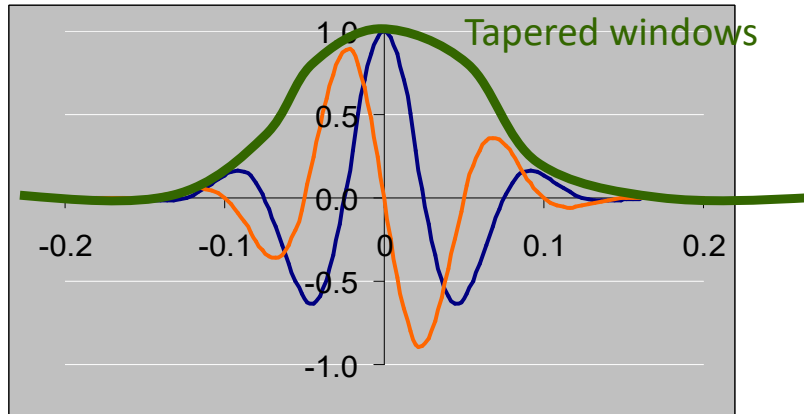
WFT spectra



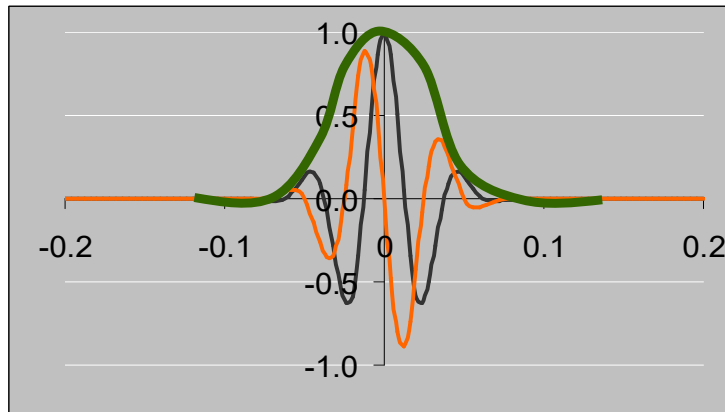
(Marfurt et al., 2015)

Morlet wavelets

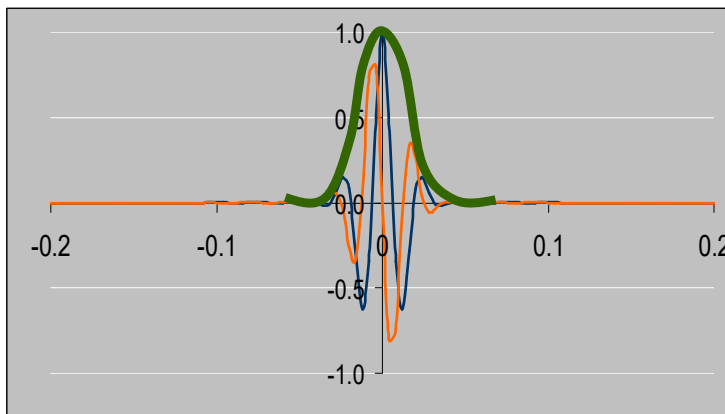
$f_c = 10$ Hz



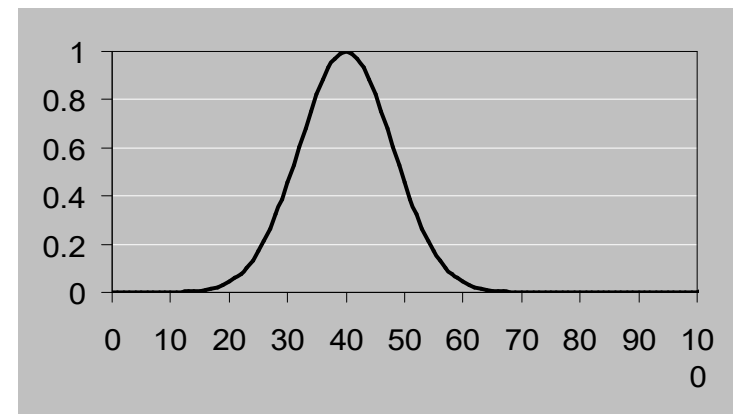
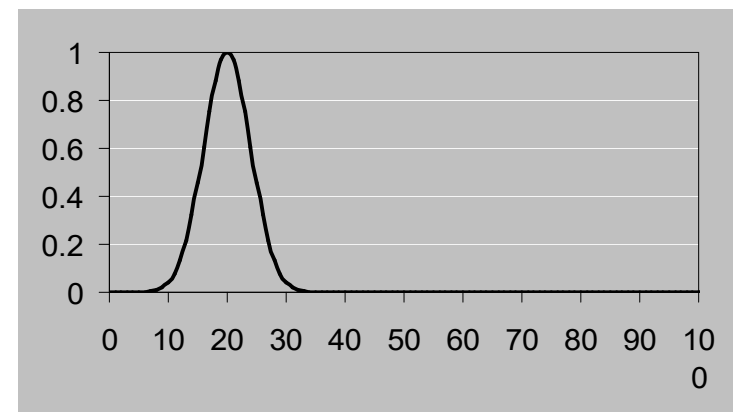
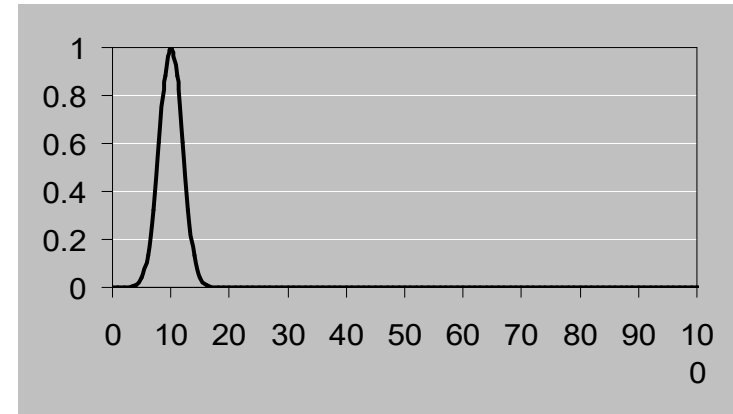
$f_c = 20$ Hz



$f_c = 40$ Hz



Morlet spectra



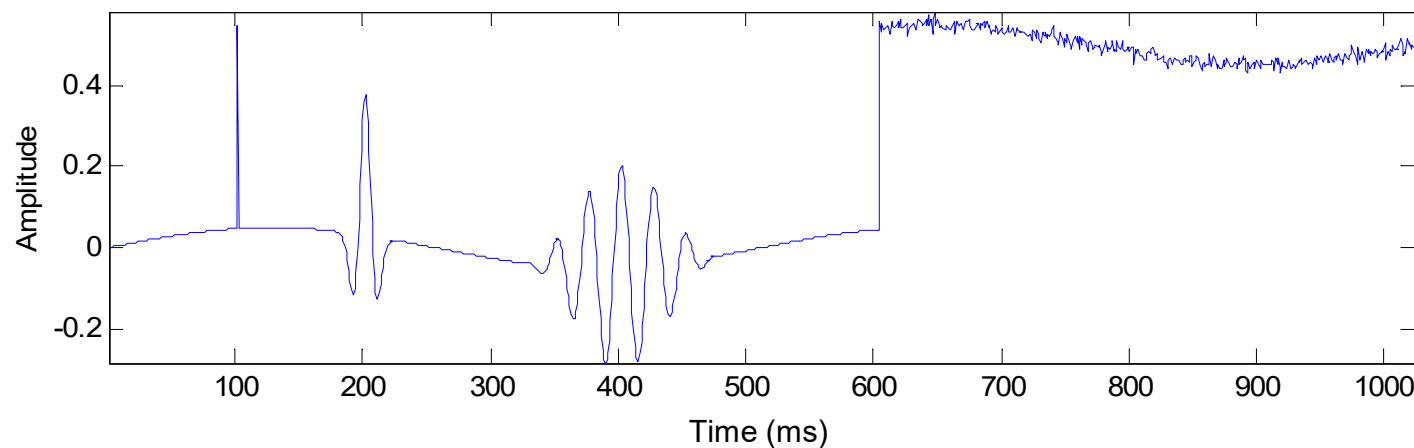
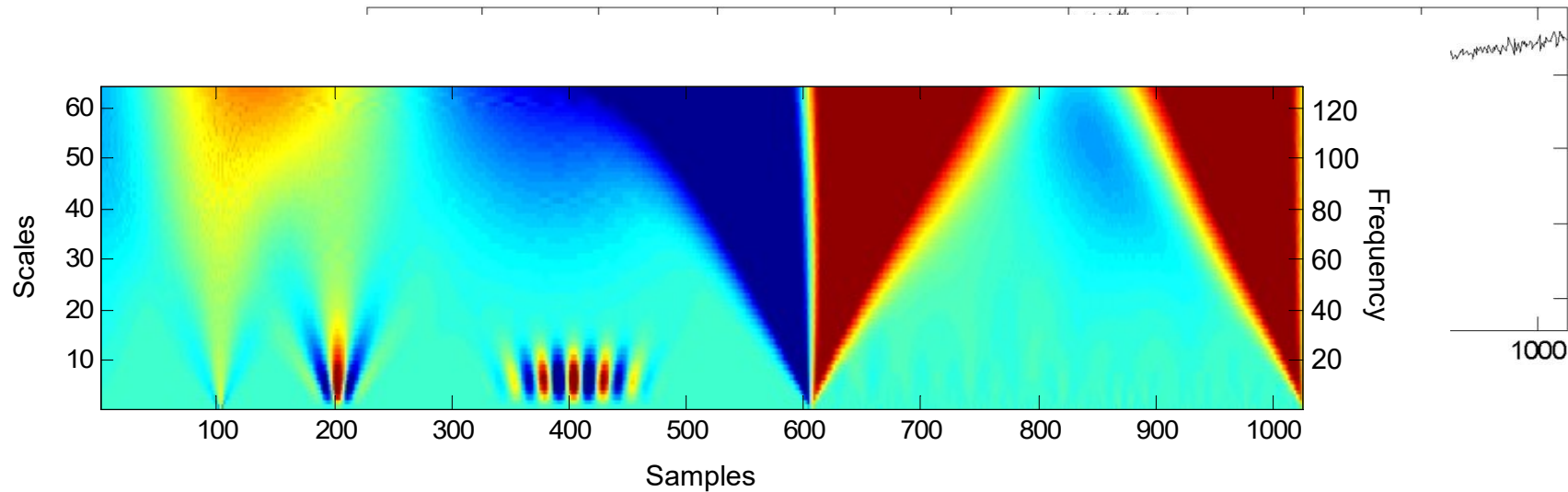
(Marfurt et al., 2015)

Continuous Wavelet Transform (CWT)

$$f(t) = \sum \alpha \psi_{\alpha}(t)$$

$$Wf(u, s) = \int_{-\infty}^{\infty} f(t) \cdot \frac{1}{\sqrt{s}} \psi^* \left(\frac{t-u}{s} \right) dt = f * \bar{\psi}_s(u) \quad \bar{\psi}_s(t) = \frac{1}{\sqrt{s}} \psi^* \left(\frac{-t}{s} \right)$$

The CWT can be interpreted as response at



(Marfurt et al., 2015)

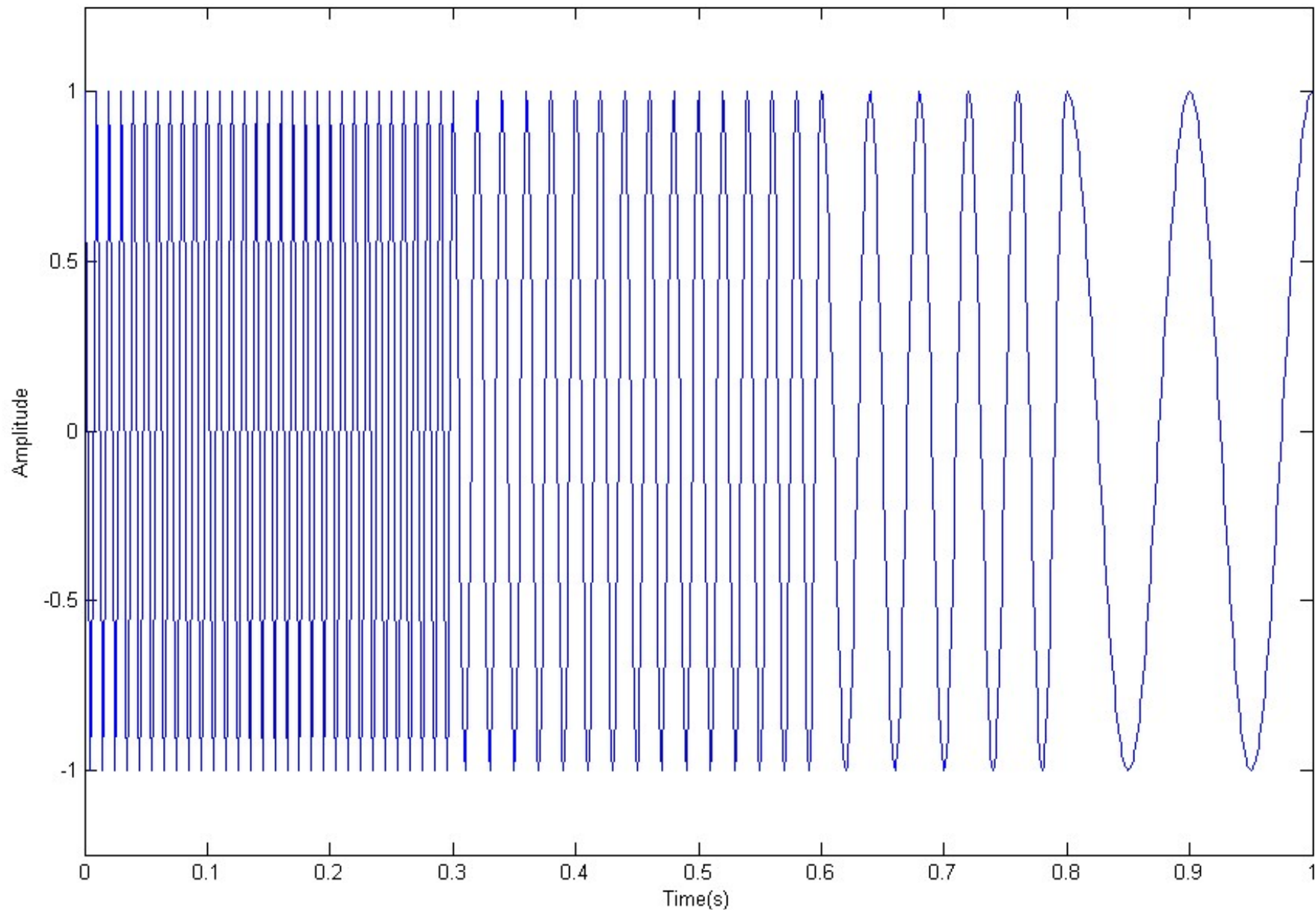


图 a, 其中0-0.3s为100Hz正弦信号, 0.3-0.6s为50Hz正弦信号,0.6-0.8s为25Hz正弦信号,0.8-1s为10Hz正弦信号.

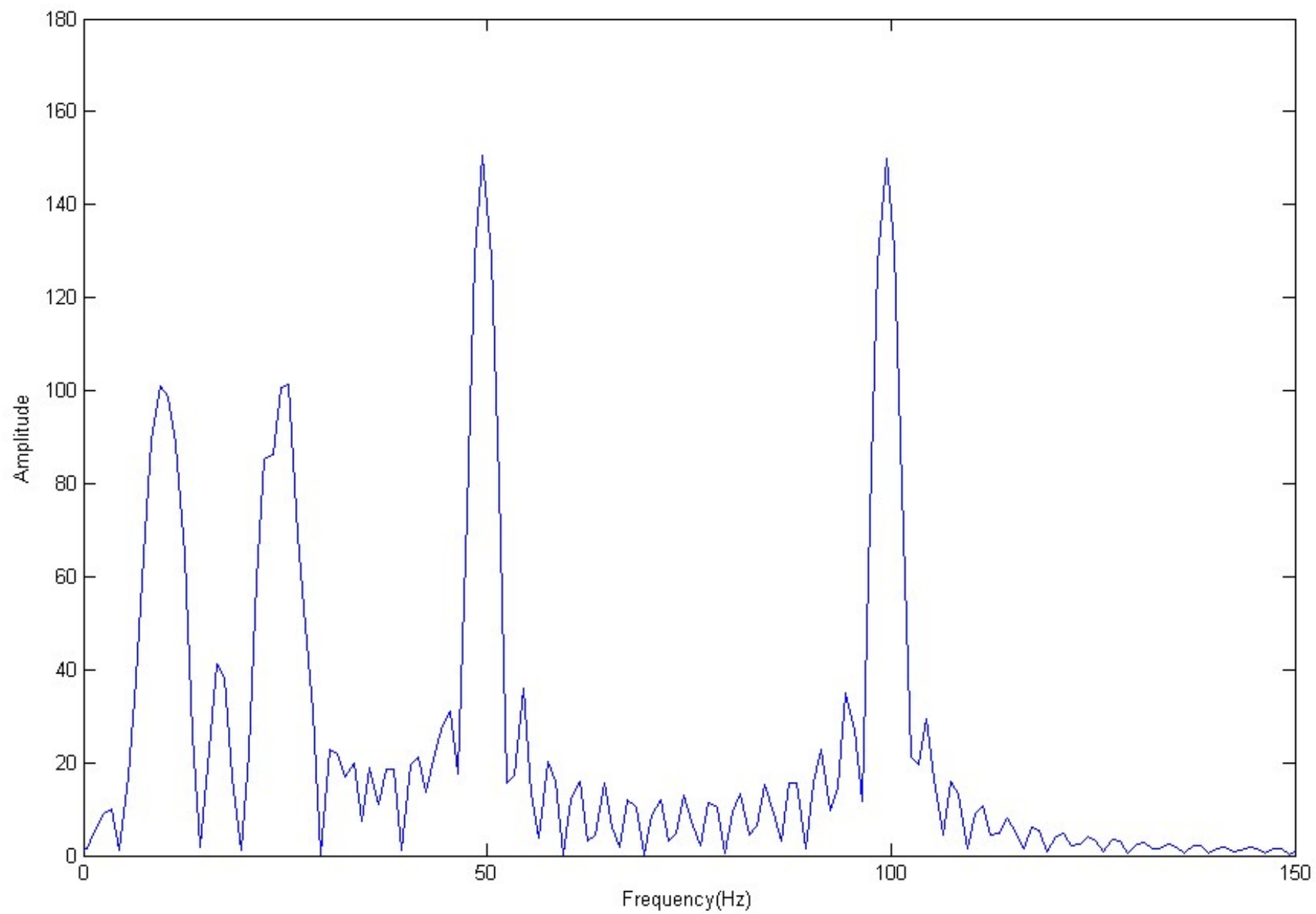


图 b, Fourier振幅谱

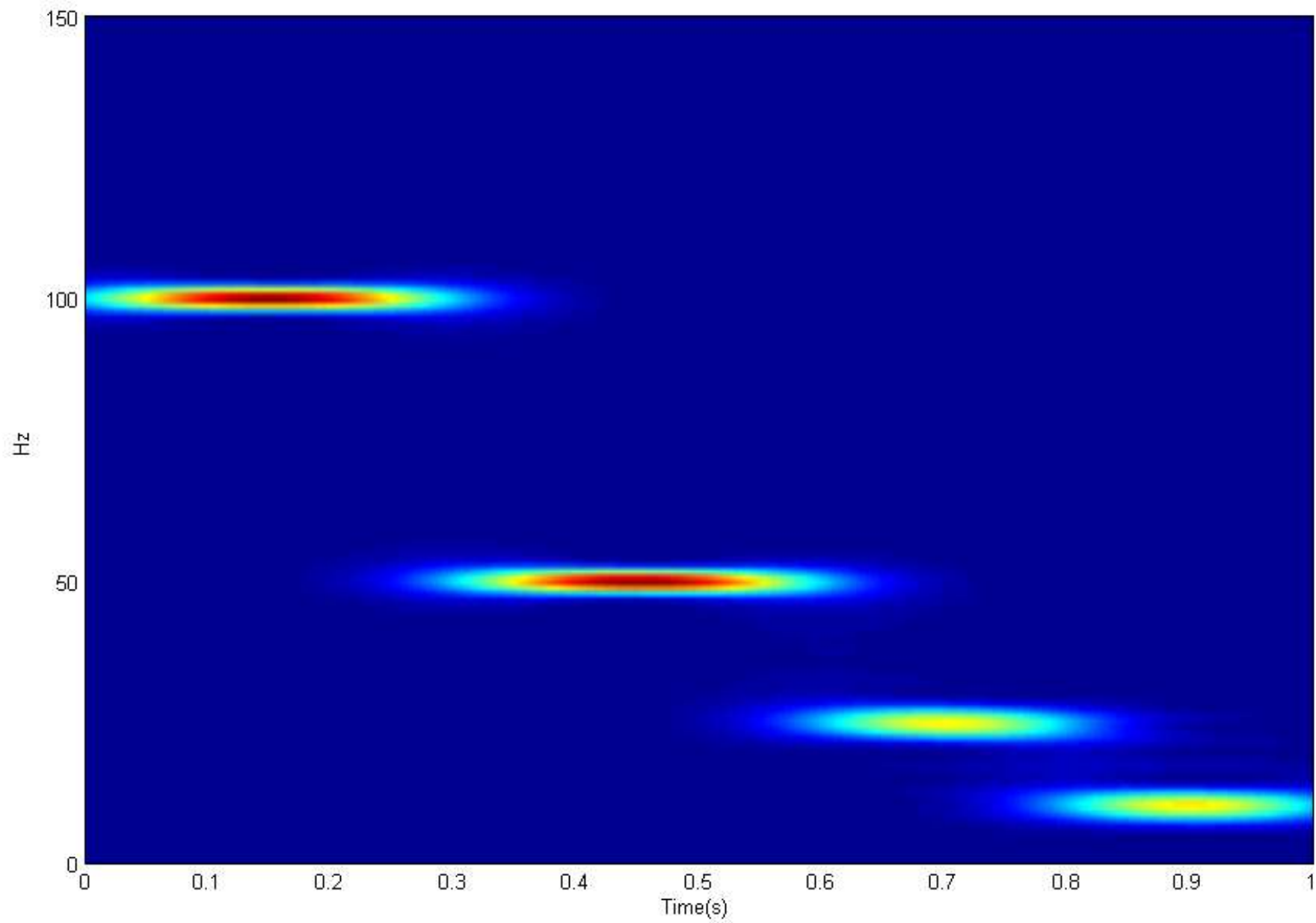


图 c, 较宽的Hanning窗对图a的短时Fourier变换

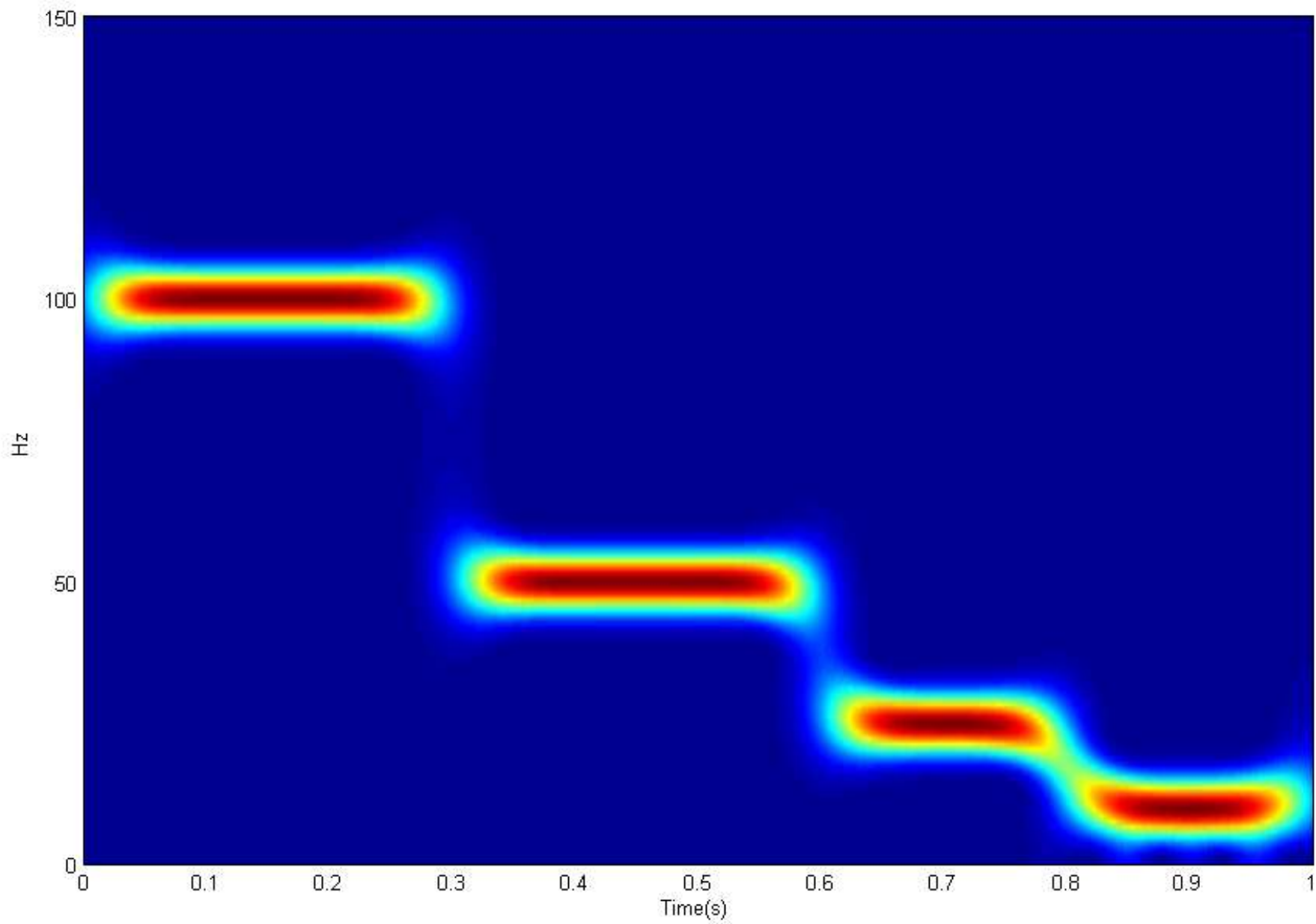


图 d, 较窄的Hanning窗对图a的短时Fourier变换

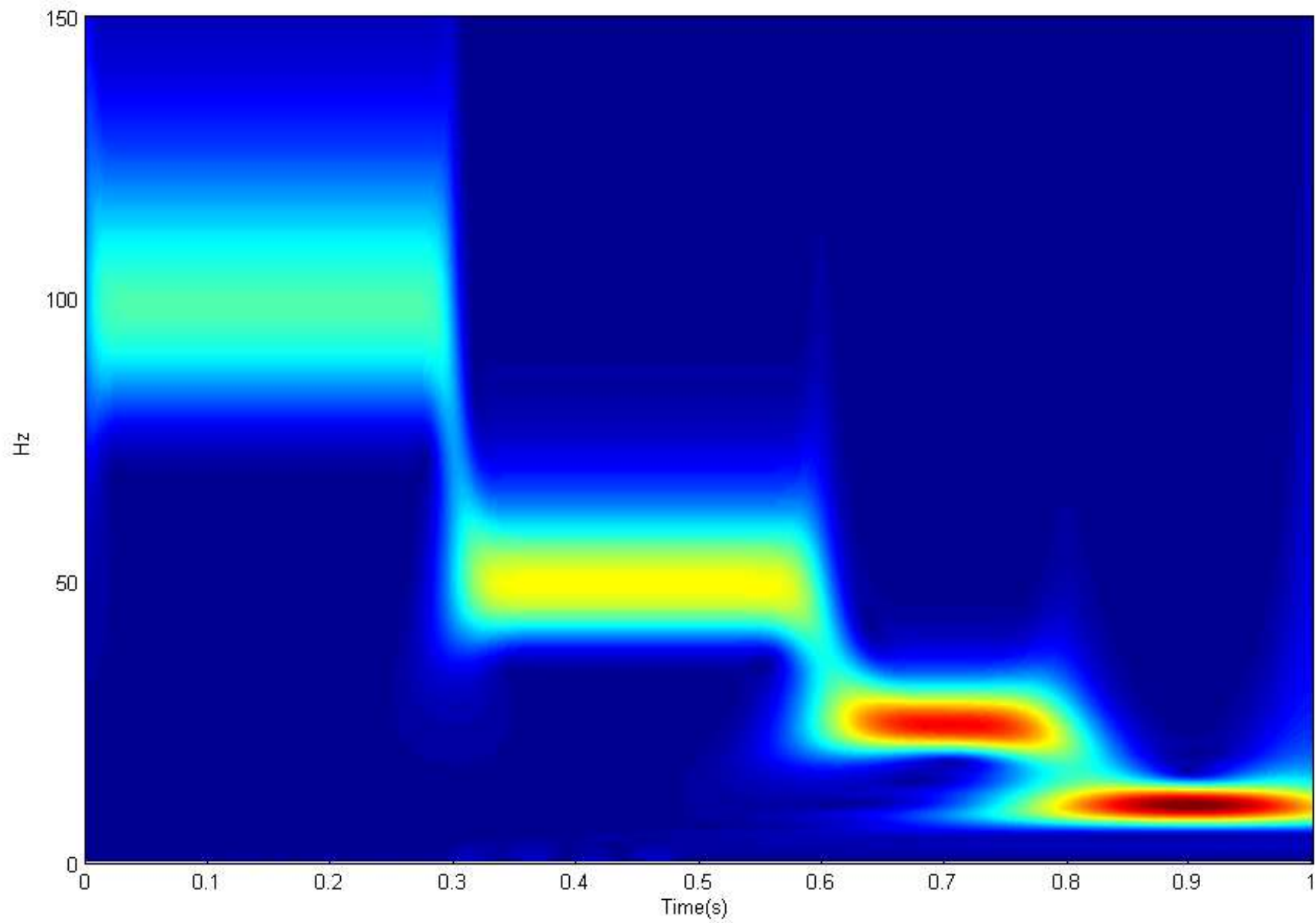
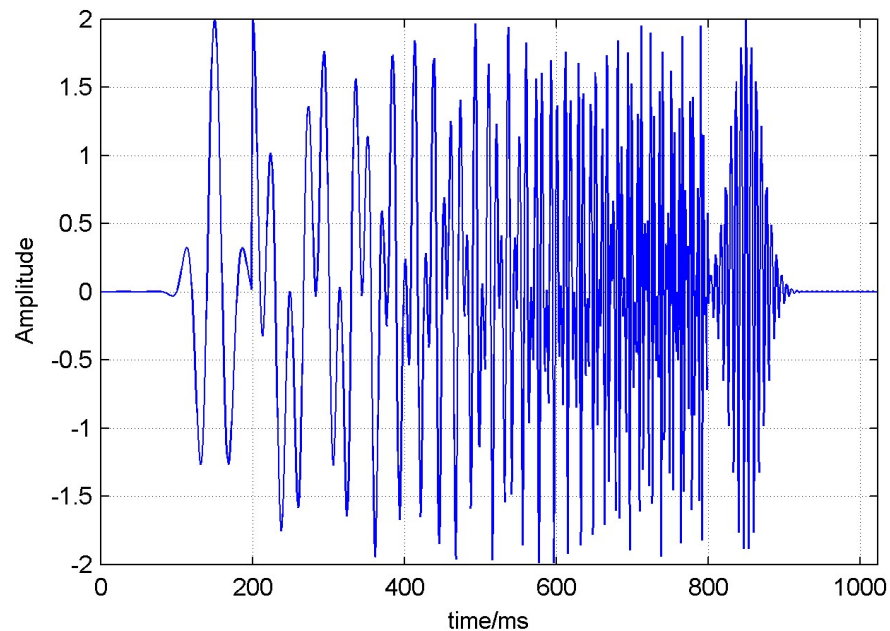
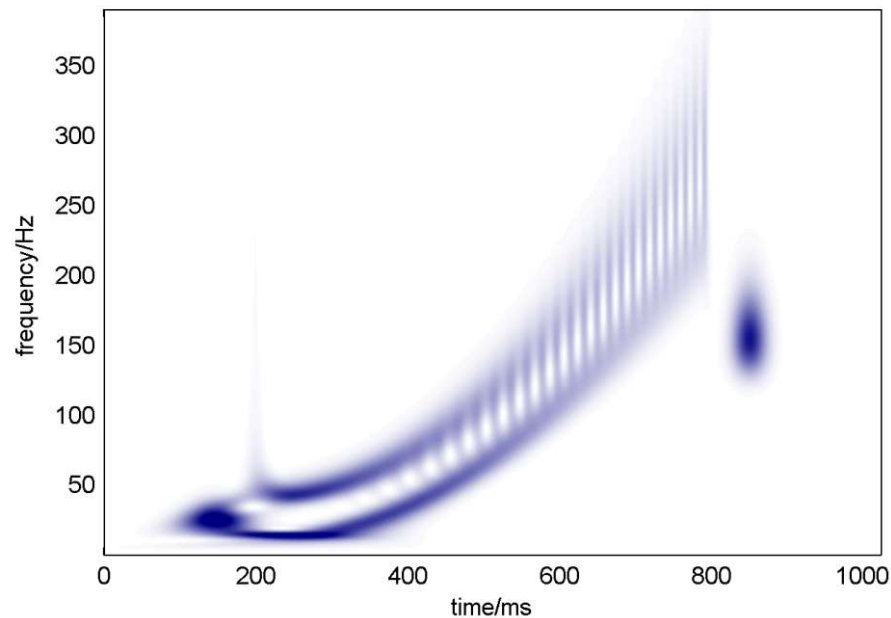


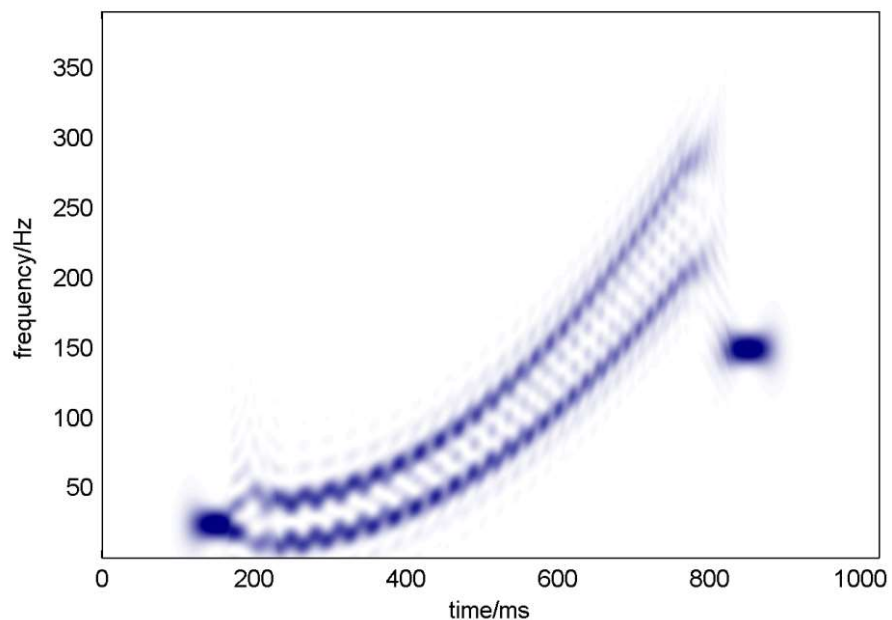
图 e, 母小波为Morlet小波对图a的小波变换



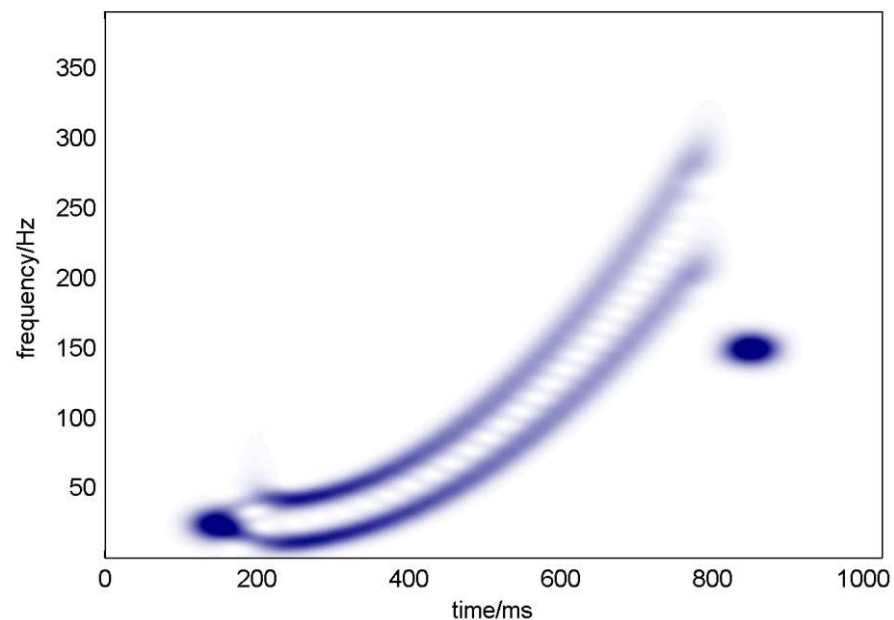
测试信号



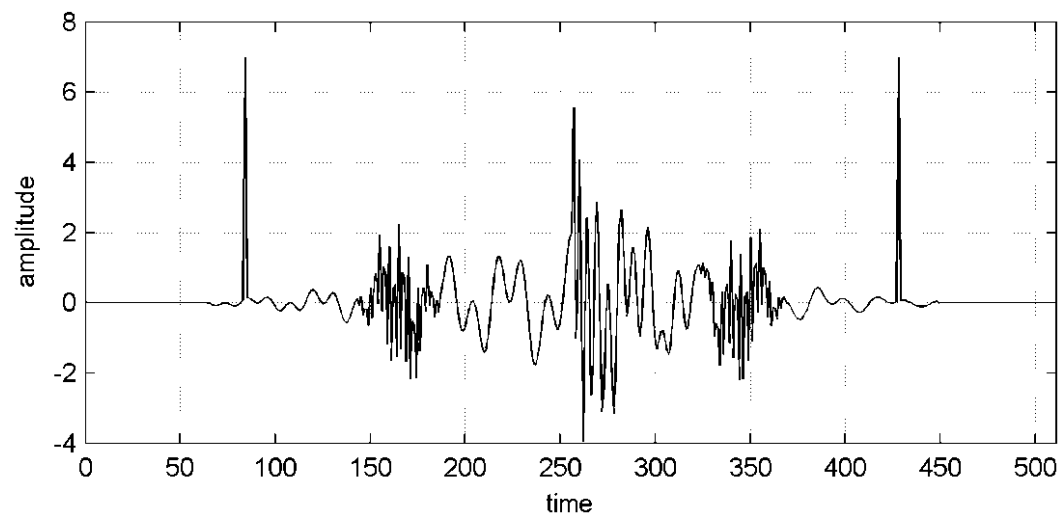
以Morlet小波为母小波的小波变换结果.



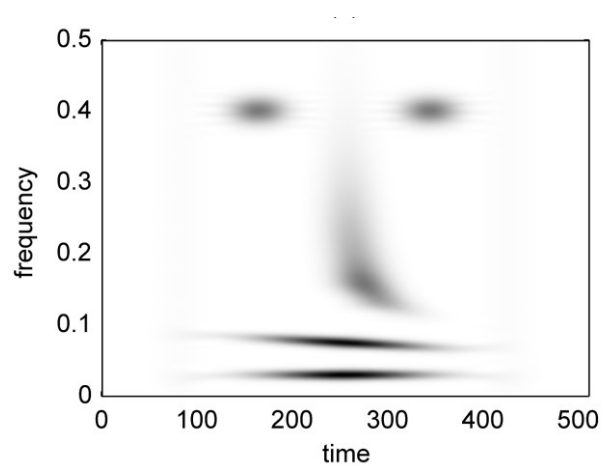
64点高斯窗短时Fourier变换结果.



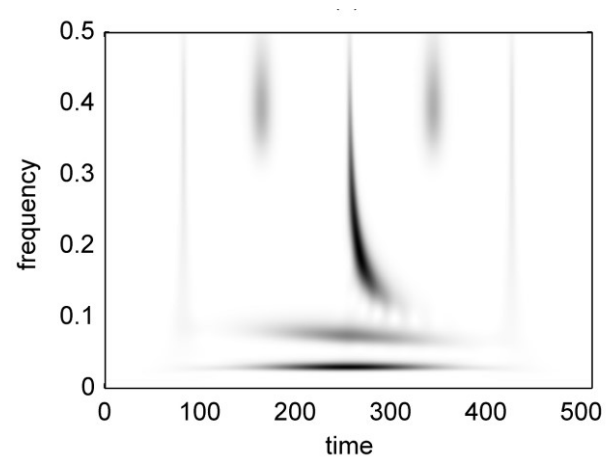
128点高斯窗短时Fourier变换结果.



测试信号

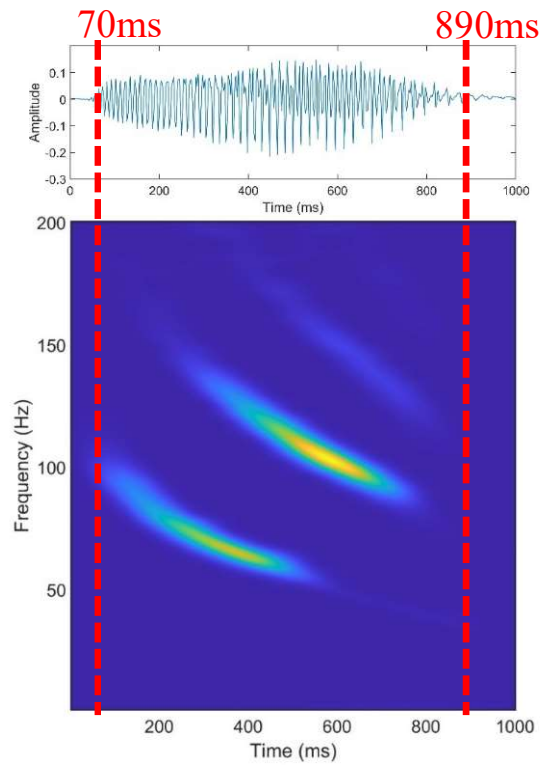


128点高斯窗短时
Fourier变换结果.



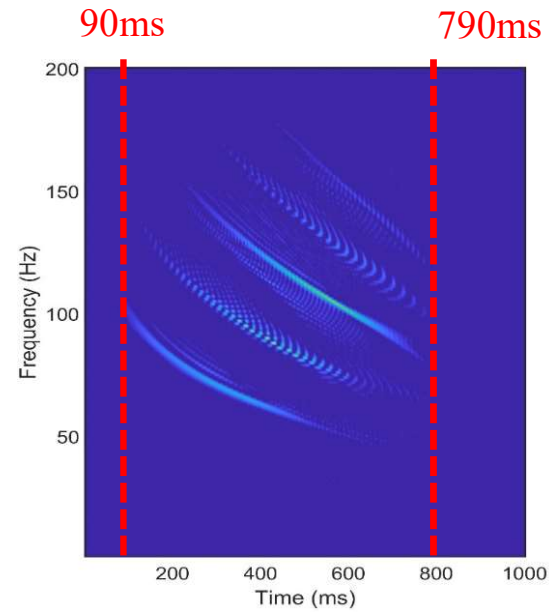
以Morlet小波为母小
波的小波变换结果.

蝙蝠回波信号

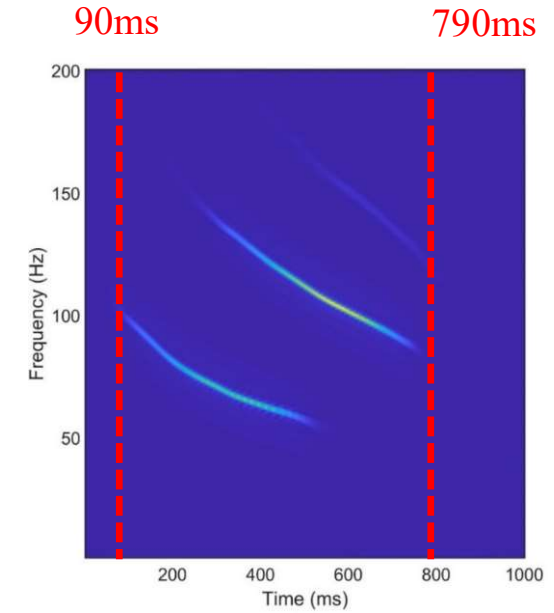


STFT谱图

采样间隔: 2.5ms
采样点数: 400

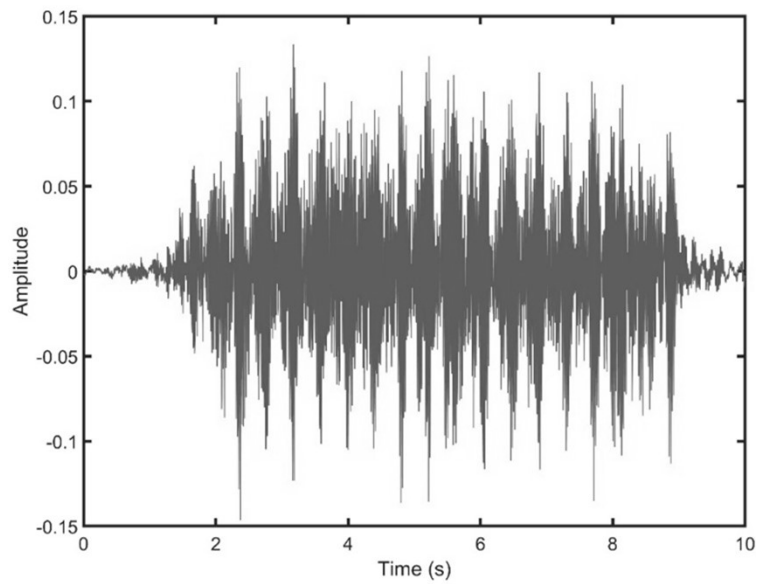


WVD

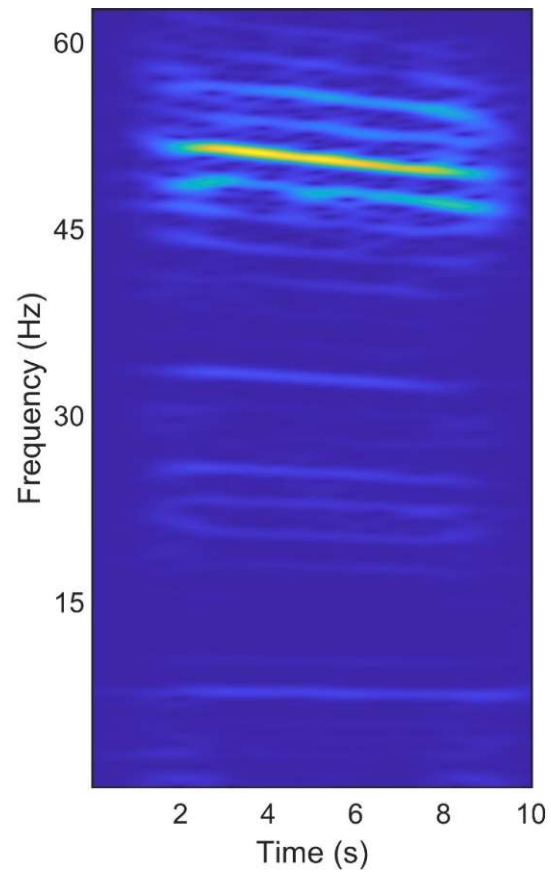


STFR-Net输出结果

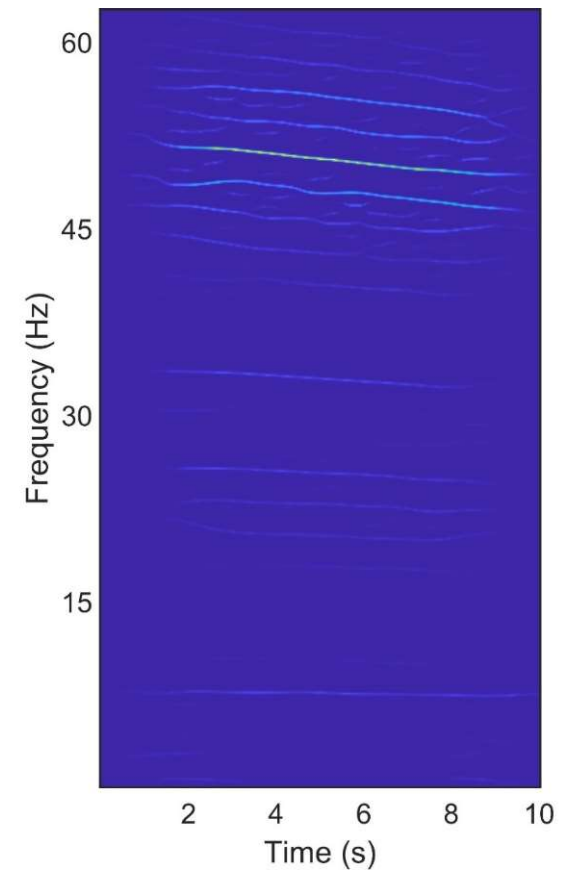
高铁震源地震信号



采样间隔: 4ms
采样点数: 2500

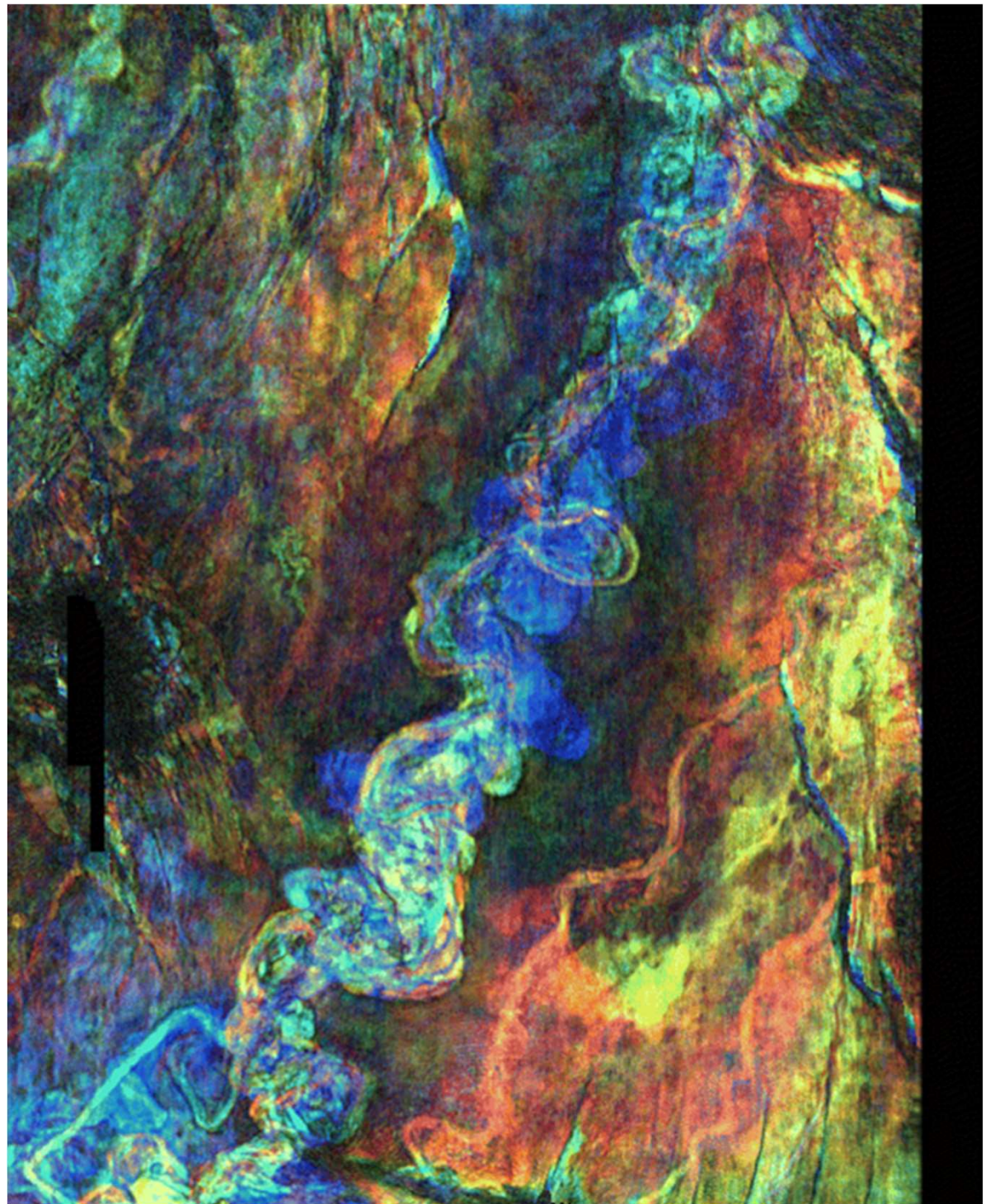


输入: STFT谱图



ISTFR-Net输出结果

频谱分解举例



(Bahorich et al., 2002)

频谱分解举例

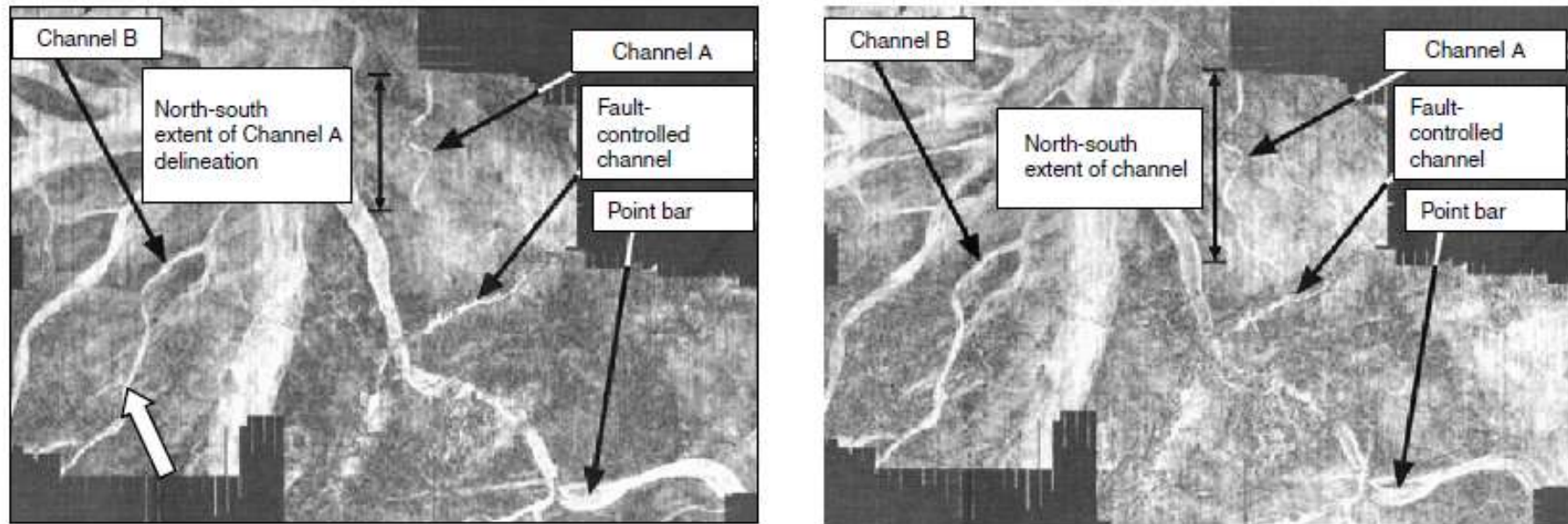
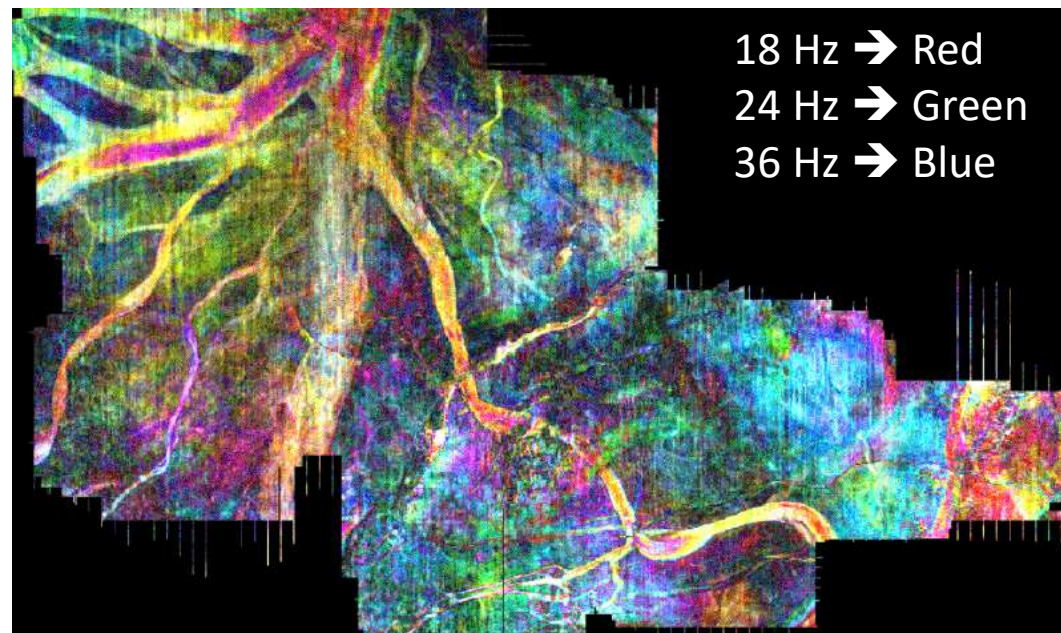


Figure 10. SWDFT spectral components at (a) 16 Hz and (b) 26 Hz along a horizon from offshore Louisiana, U.S.A., corresponding to the images in Figure 9. After Partyka et al. (1999).



(Marfurt et al., 2006)

提 纲

1. 课堂讨论课的目标及意义

2. 需要理解的内容

- 具有时间局部化性质的加窗（短时）**Fourier**分析
- 具有变时频分辨能力的小波分析、 **S**变换方法

3. 要求

要求

1. 理解加窗**Fourier**变换的物理含义；
2. 编程实现加窗**Fourier**（小波变换）正变换；
3. 利用加窗**Fourier**（小波变换）分析几种典型的合成信号，观察并分析窗函数、窗长变化对时间、频率分辨率的影响；
4. 可尝试利用加窗**Fourier**（小波变换）正变换分析实测语音等信号；
5. 整理研究、分析报告。